MILITARY SPECIFICATION

CONTROLLER, MOTOR, DIRECT CURRENT, NONMAGNETIC, WATERTIGHT, MARK 6(b), MOD 2 (MINESWEEPING)

1. SCOPE

1.1 This specification covers class HI shockproof, nonmagnetic, semi-automatic, watertight, direct current (d.c.), motor controller for Naval shipboard use. The controller (see 3.2.3) is part of an acoustic minesweeping system.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-E-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard.
MIL-C-2174 - Controllers, Direct Current, Naval Shipboard.
MIL-R-15109 - Resistors and Rheostats (Naval Shipboard).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts For Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-F-15160 - Fuses; Instrument, Power and Telephone.
MIL-M-16034/2 - Ammeters, D.C. Switchboard, 4-1/2 Inch, 250 Degree Scale.
MIL-M-16034/4 - Ammeters, A.C. Switchboard, 4-1/2 Inch, 250 Degree Scale.
MIL-I-17214 - Indicator, Permeability, Low-Mu (GO-NO-GO)
MIL-C-17588 - Circuit Breaker (Automatic-ALB) and Switch, Toggle (Circuit Breaker, Non-Automatic-NLB), Air Insulated Enclosure, 125 Volts and Below AC or DC, Naval Shipboard.
MIL-I-45208 - Inspection System Requirements.
MIL-T-55154/1 - Terminals, Feedthru (Insulated) Style FT01.
MIL-M-81203 - Manuals, Technical, In-Process Reviews, Validation and Verification Support of.

STANDARDS

MILITARY

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures For Electric and Electronic Equipment.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-470 - Maintainability Program Requirements (For Systems and Equipments).
MIL-STD-785 - Requirements for Reliability Program (For Systems and Equipments).

PUBLICATIONS

MILITARY

NAVSHIPS-0981-052-8140 - Manuals for Design of Electrical Equipment With Small Stray Magnetic Fields.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
IC 1-1965 - Industrial Control.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N.Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Ill. 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article inspection. When specified, prior to beginning production, a sample shall be examined and tested as specified in 4.2 (see 6.1).

3.2 General requirements.

3.2.1 The controller shall be suitable for starting and controlling the operation of a 5 to 7-1/2 h.p. d.c. shunt wound motor driving an acoustic mine sweeping device when used in conjunction with a dynamic braking resistor, a reverse current blocking rectifier and an automatic controller.

3.2.2 Equipment shall conform to MIL-E-917 and NEMA standards for industrial control. All components shall be items appearing on a qualified products list for their respective specification or shall be an item approved for use by the Naval Ship Engineering Center (NAVSEC).

3.2.3 The supplier may substitute static devices for magnetic devices (such as relays) providing all the required functions are performed, proof is shown that the supplier has manufactured approved static control devices previously, there is no increase in size or weight, and an equivalent delivery time can be met.

3.3 Design characteristics.

3.3.1 Materials. Nonmagnetic materials shall be used for all parts except those that must be magnetic to perform their function. Enclosures shall be made of nonmagnetic corrosion-resisting steel. Nonmagnetic is defined as materials having a permeability not exceeding 2.0 after fabrication. All possible effort shall be made to minimize the presence of strong fields. Design guide lines may be obtained from NAVSHIPS 0981-052-8140.

3.3.2 Operation. Manual operation of a pushbutton shall actuate magnetic componets as required to start a shunt wound motor.

3.3.2.1 Starting shall be accomplished by the conventional means of starting resistors and accelerating contactors. Controller shall provide low voltage protection (LVP). Controller shall also have a pilot relay circuit which will operate when the motor armature current reverses and will close contacts to shunt the brake resistor across the armature.

3.3.3 Components. The following components shall be included in the controller and wired into the circuit as required:

(a) Current sensitive overload relay with reset button in accordance with MIL-C-2174. Locate on the controller control panel.
(b) A 0 to 100 amp d.c. ammeter in accordance with MIL-M-16034/4 with 50 millivolt (mv) external shunt to indicate motor armature current. Locate meter on controller control panel and shunt within controller. Provide calibrated leads.
(c) A 0 to 1 ma a.c. milliammeter in accordance with MIL-M-16034/2 calibrated as a speed indicator. The meter face shall have two scales. One labeled 0 to 40 cps (full scale with 40 divisions and one labeled 0 to 2400 revolutions per minute (r.p.m.) (full scale with 48 divisions). Milliammeter to be calibrated and modified by contractor. Calibrate dial from 60 cycle supply at 20°C. Each 50 r.p.m. graduation shall be equal to 1.580 volts root mean square (r.m.s.). Locate meter on controller control panel.

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(d) A 3-pole, 50 amp, 120 volt d.c. air circuit breaker in accordance with MIL-C-17588 for main circuit breaker. Locate on controller control panel.
(e) Start pushbutton in accordance with MIL-C-2174. Locate on controller control panel.
(f) Stop pushbutton in accordance with MIL-C-2174. Locate on controller control panel.
(g) Indicating light (white) in accordance with MIL-C-2174 to indicate power available to controller. Locate on controller control panel.
(h) Indicating light (red) in accordance with MIL-C-2174 to indicate water leaking into the remote acoustic device. This light to be actuated by flood relay (see j). Locate on controller control panel.
(i) 115 volts d.c. relay to actuate the red indicator light and with a contact wired in series with the stop pushbutton capable of returning the system to stop. Locate in controller.
(j) Fuses shall conform to MIL-F-15160.
(k) Terminal boards shall conform to MIL-T-55164/1.

3.3.4 Auxiliary function. The following auxiliary function shall be provided:

(a) A means of preventing chatter of the flood relay due to a.c. pulses received from the tachometer generator in the acoustic device.

3.3.5 Interface provisions. Terminal blocks shall be provided in the controller to allow interconnecting with the other system components as follows:

(a) 2 terminals for incoming 115 volts d.c. power (designated "LINE +" and "LINE-").
(b) 2 terminals for connection to the motor driven rheostat in the automatic controller AMK4 (V) NM-AMK6 (b) NM. (Each terminal shall be designated "FIELD INTLK".)
(c) 2 terminals for connection to inverse current blocking rectifier (designated "RECT +" and "RECT -").
(d) 2 terminals for connection to dynamic braking resistor (each designated "BRK RES").
(e) Cable entrance plates in accordance with MIL-E-2036 shall be provided in the bottom of the controller.

3.3.5.1 A Crouse-Hinds Model No. 6728 receptacle or equal shall be provided on bottom of controller to mate with plug on connecting cable between controller and A Mark 6(b) type acoustic device. The connecting cable will connect the terminals of the Crouse-Hinds receptacle in the controller to the terminals of the motor in the A Mark 6(b) type acoustic device (see table I).

Table I - Connecting cable terminals.

Color code of Crouse-Hinds terminal	Motor terminal
White	Common armature and field
Black	Armature
Green	Field
Red	Flood switch and tachometer generator

3.3.6 Grounding. Equipment shall be designed for operation on an ungrounded system but shall operate satisfactorily with an accidential ground on any power conductor.

3.3.7 Simplification, standardization and compactness. Controllers shall be simplified and standardized to the maximum extent, consistent with obtaining reliable operation and the required operating features.

3.3.8 Overload protection. Protection shall be provided during both starting and running conditions.

3.3.9 Ambient temperature. Equipment shall be designed for a 50°C. ambient temperature. Overload relay shall be calibrated for 40°C. ambient temperature.

3.3.10 Temperature rise. The temperature rise of the various components within an assemblied controller shall not exceed the values shown in MIL-C-2174.

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3.3.11 Wiring. Permanent internal wiring shall be copper wire or bus bar. The size, stranding, and insulation of wire, and the dimensions, insulation, and spacing of bus bars shall be mechanically and electrically suited to the application. Wire not exposed to heat shall have thermoplastic insulation rated for an operating temperature of 105°C. Connections to resistors or class H coils, and wires otherwise exposed to heat, shall be insulated with thermoplastic, asbestos and an outer braid. The minimum wire size shall be No. 14 American Wire Gage (AWG).

3.3.12 Voltage variation. Controllers shall be designed to operate between 90 and 130 volts d.c. at 50°C. ambient temperature.

3.3.13 Inclination. Equipment shall operate at rated voltage when inclined to an angle of 45 degrees from the vertical in any direction and not malfunction when inclined to an angle of 60 degrees from the vertical in any direction.

3.3.14 Vibration. Equipment shall be capable of withstanding the type I vibration tests of MIL-STD-167 without mechanical damage, contact chatter, or other malfunctioning.

3.3.15 Shock. The completed controller shall be capable of withstanding the shock tests specified in MIL-S-901 for class I grade A equipment without the use of shock mounts or manual latches. With rated voltage available, the load shall not be energized or deenergized because of shock except that line contacts may open momentarily.

3.3.16 Insulation resistance. Insulation resistance, after completion of electrical tests, shall be not less than 10 megohms.

3.3.17 Dielectric strength. Equipment shall be capable of withstanding dielectric tests as specified in NEMA ICl-2.42.

3.3.18 Arc protection. Interior surfaces of metallic enclosures which are in the wake of electric arcs shall be protected by asbestos or glass melamine sheets.

3.3.19 Enclosure. The enclosure shall be watertight in accordance with NEMA ICl. Enclosure shall have a moisture drain in the bottom. The opening shall be closed with a 1/4 inch or larger pipe plug. Enclosure shall comply with MIL-STD-108.

3.3.20 Connection diagram and table of overload heaters. Each controller shall include a connection diagram, a schematic diagram, a description of operation and a table of overload heaters or coils. This information shall be printed by a process that is non-fading, protected by transparent plastic and secured to the inside of the enclosure.

3.3.21 Identification and information plates. Each equipment shall be provided with an identification plate containing the manufacturer's name, year of manufacturer, weight, contract number, and Federal stock number. Controller identification plates shall also contain the voltage and maximum horsepower rating. The preferred location is outside the enclosure.

3.3.22 Size. Overall size of the controller (not including mounting tabs or brackets) shall not exceed 31-5/8 inches in height, 17-3/4 inches in width and 19 inches in depth. Four mounting holes shall be arranged to have centerlines at the corners of a rectangle 19-3/4 inches wide and 20 inches high.

3.3.23 Weight. The total weight of the controller shall not exceed 200 pounds.

3.4 Repair parts. Each repair part shall be equal to the original and suitable for replacement without special tools, and without the removal and replacement of rivets or pins dependent upon a force fit. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.4.1 Quantity of repair parts. Repair parts, furnished with controllers, shall be furnished in sets as follows. The number of sets shall be specified (see 6.1):

One set of repair parts:

Complete replacement of contacts
Complete replacement of springs
Complete replacement of overload heaters

Shunt coils, one of each size and type
Resistors, one of each size and type
Capacitors, one of each size and type

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3.4.2 Standard items. Repair parts shall not be furnished for Navy standard items, such as fuses and indicator lights. Repair parts shall be furnished for proprietary items.

3.4.3 Overload relays. No springs, contacts or copper series coils shall be furnished for overload relays or associated devices.

3.4.4 Coils. The quantity of each size and type of shunt coil shall be not less than 10 percent of the quantity installed in one controller.

3.4.5 Resistors. The quantity of each type shall be not less than 10 percent of the quantity installed.

3.4.6 Provisioning technical documentation shall be in accordance with MIL-P-15137.

3.5 Drawings. Drawings shall conform to MIL-D-1000 and MIL-D-1000/2 and shall contain the following minimum data:

(a) List of material
(b) Drawing of the front top and side views of the controller, giving dimensions and essential details of the enclosure, including quantity and size of mounting bolts required. The drawing shall also show the outline of the components keying them to the bill of material.
(c) Connection and schematic diagrams.
(d) Description of operation of the controller.
(e) Description of adjustments which can be made on the equipment.
(f) Table of overload heater or coil catalog numbers, with motor full load current ranges.
(g) Table of applicable shunt coil catalog numbers with corresponding voltage, ohms, turns, wire size and covering, coil insulation and impregnation, and device on which used. This information is not required for encapsulated coils.
(h) Table of applicable resistor catalog numbers.
(i) Capacitor data.
(j) List of exceptions to this specification.
(k) Descriptive data of the equipment which shall include rating and catalog number.
(l) Weight of the equipment and an unboxed set of repair parts.
(m) List of repair parts with piece numbers for each item, the name of each item, the quantity of each item recommended for one set of repair parts and the manufacturer's identification for the item.
(n) Title block.

3.6 Technical manuals. Prior to production, drawings shall be submitted to NAVSEC for approval. When required (see 6.1), manuals (manuscript only), including reproducible art work, shall be prepared containing information necessary for operation and maintenance of the controllers. Parts identification shall be included and shall contain brief descriptive information for basic and repair parts to facilitate identification of the parts in the assembly, and to permit ordering of replacement parts. Graphics used to supplement the descriptive matter contained in the text for parts identification shall consist of photographs or line drawings or both. Color shall not be employed. Appendix I and Appendix II, respectively, are applicable to controllers previously procured and are furnished merely to illustrate the type of information required in the manuals; they are not intended to indicate details of construction required by this specification. The text of the manuscript should be arranged in the format shown.

3.6.1 Preliminary section. The contractor shall forward two copies of the manuscript in preliminary form, together with two copies of each photograph or line drawing required to illustrate the text to the procuring activity for approval. They shall be submitted by a letter of transmittal via the cognizant Government inspector.

3.6.2 Final section. Upon approval of the preliminary section three copies of the final section shall be forwarded via the cognizant Government inspector to the procuring activity for final approval. The final section shall be similar to the preliminary section except that all changes to the contents made by the procuring activity shall be incorporated therein. Three reproducible copies of each photograph or line drawing required to illustrate the text shall be furnished with the final section. Insofar as possible the illustrations shall not exceed 8 x 10 inches. After final approval, two copies of the final section, together with two copies each of the reproducible photographs or line drawings shall be forwarded by the procuring activity to the Naval Ship Engineering Center, Washington, D.C. 20360. Naval Ship Systems Command (NAVSHIPS) and section numbers will be assigned to this section by this activity. Maintenance procedures shall be validated and verified in accordance with MIL-M-81203 before the equipment is delivered.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain shall be in accordance with MIL-I-45208. Reliability and maintainability programs in accordance with MIL-STD-785 and MIL-STD-470 shall be established and in effect during the accomplishment of contracts under this specification.

4.2 First article inspection. First article inspection shall consist of the examination and tests specified in table II. Report of the tests should be forwarded to the procuring activity concerned for approval. The cognizant DCAS office shall be advised as to the date when tests shall be performed.

Table II - Test agenda.

Description and order of examinations and tests	First article inspection	Routine inspection	Applicable paragraph	
			Requirement	Examination and test
Workmanship, fit of parts and materials	X	X	3.3.1	4.5.1
Weight	X		3.3.23	4.5.2
Creepage and clearance	X		-	4.5.3
Resistance	X	X	-	4.5.4
General operation	X	X	-	4.5.5
Operation with acoustic minesweeping device	X		3.2.1	4.5.6
Inclination	X		3.3.13	4.5.7
Voltage variation	X	X	3.3.12	4.5.8
Temperature rise	X		3.3.10	4.5.9
Insulation resistance	X	X	3.3.16	4.5.10
Dielectric strength	X	X	3.3.17	4.5.11
Shock	X		3.3.15	4.5.12
Effectiveness of enclosure	X		3.3.19	4.5.13

4.2.1 A controller which has successfully passed the first article inspection may be submitted for inspection as production equipment under the contract.

4.3 There shall be no change between first aricle sample and production models unless approved by the Government.

4.4 Comparison with approved first article sample (see 6.1). The Government representative shall review the first article report and shall ascertain whether the production controllers are of the same design and construction as the approved first article sample. Submission of production items on a new contract or order shall be accompanied by a statement from the manufacturer certifying that there has been no change in design, material, or construction from that of the approved first article samples.

4.5 First article and routine tests (tests made on every production unit) shall be in accordance with table II.

4.5.1 Equipment examination. Each equipment shall be given an examination to determine that the design, material and workmanship conform to this specification. The fit of parts shall be observed, with particular reference to interchangeability of replaceable parts. Conformance to approved drawings, with respect to material, finish, construction, dimensions and weight, shall be verified. This examination should not include disassembly to the point that performance, durability or appearance would be affected.

4.5.1.1 All applicable components of the controller shall be checked with a permeability indicator conforming to MIL-I-17214.

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4.5.1.2 Repair parts. All repair parts shall be subjected to an examination to ascertain that the materials, workmanship and finish are first class in every respect and that they conform fully to the manufacturer's approved drawings.

4.5.2 Weight. The weight of the equipment, and the weight of the repair parts, shall be taken and recorded. The weight shall not differ from the weight specified on the drawing of the equipment by more than 10 percent.

4.5.3 Creepage and clearance. Creepage and clearance distance shall be demonstrated by actual measurement to be in accordance with MIL-E-917.

4.5.4 Resistance. The overall resistance of shunt coils and resistors shall be measured to determine that, when corrected for differences in temperature, they are within plus or minus 10 percent of the design values.

4.5.5 General operation. General operation includes all tests necessary to ascertain that the required operating features function proplerly. In particular, it shall be determined that the required sequence of operation has been attained and that low-voltage protective devices and accelerating contactors operate as required.

4.5.6 Operation with acoustic minesweeping device. When specified (see 6.1) the first article sample with associated repair parts shall be submitted to the U.S. Navy Mine Defense Laboratory, Panama City, Florida 32402, for operational tests in which the manual controller together with a dynamic braking resistor, reverse current blocking rectifier and automatic controller will be used to start and control the operation of a 5 to 7-1/2 HP d.c. shunt wound motor driving an acoustic minesweeping device.

4.5.7 Inclination.

4.5.7.1 Forty-five degree inclination. Equipment shall be tested for proper operati at rated voltage and current, when inclined 45 degrees forward, backward and to each side.

4.5.7.2 Sixty degree inclination. Equipment shall be tested in the deenergized condition, and when energized at rated voltage and current, to determine that contacts do not open or close from their intended position when inclined 60 degrees forward, backward and to each side.

4.5.8 Voltage variation. Contactors, relays and other devices using shunt coils shall be tested to insure conformance with 3.3.12.

4.5.9 Temperature rise. The test methods to be employed and the precautions to be observed in conducting temperature rise tests shall be in accordance with NEMA IC 1-2.41, 1-13.40, 1-13.41 and as specified in 4.5.9.1 and 4.5.9.2.

4.5.9.1 Temperature rise tests shall be made at rated voltage, rated load and the duty specified, to determine that the temperature rises specified in 3.3.10 are not exceeded. Temperature rise tests shall be made on individual parts of the completely assembled equipment. The specified current or voltage shall be applied to all parts simultaneously so that the mutual heating effect of the several parts will be present. Temperature rise tests shall be made with enclosures in position.

4.5.9.2 The first time a controller is presented for inspection, temperature rise tests shall be made on the completely assembled controller. After a specific controller has passed temperature rise tests, these tests are not required on identical controllers delivered at a later date.

4.5.10 Insulation resistance. Insulation resistance shall be as specified in 3.3.16. Insulation resistance shall be measured with an insulation-resistance-indicating meter with full scale reading of 100 or 200 megohms, an open circuit voltage of 500 volts and a voltage of at least 450 volts across a resistance of 1 megohm. The measurement of insulation resistance shall be made successively between each circuit and all other circuits and metal parts grounded. The temperature and relative humidity shall be measured and recorded.

4.5.11 Dielectric strength tests. Dielectric strength tests shall be conducted in accordance with 3.3.17. The tests shall be made upon the completely assembled equipment after completion of all other tests except the shock test.

4.5.12 Shock tests. Shock tests shall be conducted in accordance with 3.3.15 and as specified in 4.5.12.1 and 4.5.12.2.

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4.5.12.1 Method of mounting. Use figure 4A of MIL-S-901. Equipment shall be mounted on the shock machine in a manner simulating normal shipboard mounting.

4.5.12.2 Definition of failure. Breakage or appreciable distortion of any parts, including mounting bolts. Electrical malfunction such as starting up if deenergized, stopping or recycling if energized, loss of insulation resistance or dielectric strength.

4.5.13 Effectiveness of enclosure. Enclosures shall be tested to determine compliance with 3.3.19.

4.6 Inspection of preparation for delivery. Packaging, packing and marking shall be inspected to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts. (see 5.3)

5.1.1 Controllers.

5.1.1.1 Preservation and packaging. Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1.1.3 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be packaged in heat sealable laminated paper and foil bags or in heat sealable plastic film. When a complete switch is furnished as a repair part and the switch is provided with a watertight or submersible enclosure, the paper and foil bag, or plastic film, shall not be required.

5.2 Domestic shipment and storage or overseas shipment. The requirements and levels of preservation, packaging, packing and marking for shipment shall be specified by the procuring activity (see 6.2).

5.3 Use of polystyrene (loose-fill) material (see 6.1).

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.2), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

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5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Number of sets or repair parts required (see 3.4.1).
(c) Technical manual, if required (see 3.6).
(d) First article sample if required (see 4.4 and 6.3).
(e) Preservation, packaging, packing and marking if other than 5.1 (see 5.2).
(f) Whether the use of polystyrene is authorized (see 5.3).
(g) Quantity required.

6.2 First article inspection. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.3 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.4 Items not covered by this specification. The other components of the system are:

(a) A switchboard that supplies 115 volts d.c. to the manual controller.
(b) A resistor assembly (5.22 ohms) used in conjunction with the dynamic braking circuit.
(c) A rectifier assembly used for blocking inverse armature current.
(d) An automatic controller which provides a varying resistance via a motor driven rheostat. This rehostat is connected in series with the field of the motor described in (e) to vary the motor speed.
(e) An acoustic minesweeping device containing a 5 to 7-1/2 (h.p.), 115 volts, 660 to 2070 r.p.m., 41 to 60 amp. shunt wound motor, an a.c. generator to provide input to a tachometer indicator in the manual controller, and a flood switch that will close and signal the controller if the acoustic device is leaking. The manual controller is interconnected with the resistor assembly, the rectifier assembly, the automatic controller and the acoustic minesweeping device. The acoustic device is equipped with a flywheel which helps to provide a smooth variation in motor speed during the buildup portion of the cycle. When the motor field is strengthened to decrease the speed, the motor acts as a generator due to the flywheel inertia and the armature current reverses and tends to be fed back into the line. The rectifier prevents this, and by means of a pilot relay circuit and contacts in the manual controller the brake resistor is shunted across the armature, absorbs the reverse current and acts as a load on the motor, slowing it down to base speed.

Preparing activity:
Navy - SH
(Project 1075-N014)

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APPENDIX I

This appendix is an example of the type of information required in the Technical Manual for operation of the manual controller.

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OPERATION AND MAINTENANCE OF MINE COUNTERMEASURES EQUIPMENT

Part II. Operation of Mine Countermeasures Gear and Auxiliary Equipment

Chapter 13. Acoustic Minesweeping Gear

SECTION 13-224

MANUAL CONTROL ASSEMBLY, A MARK 6(b) NM MOD 1

(MANUFACTURED BY MECO INC.)

NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING
PRINCE GEORGE'S CENTER
HYATTSVILLE, MARYLAND 20782

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RECORD OF CHANGES AND CORRECTIONS

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Part 1 - Introduction

1-1. SAFETY PRECAUTIONS

1-1.1 General.

(a) The attention of officers and operating personnel is directed to chapter 18 of the United States Navy Safety Precautions (OPNAV 34 P1) or superseding instructions on the subject of safety precautions to be observed.

(b) This equipment employs voltages which are dangerous and may be fatal if contacted by operating personnel. Extreme caution should be exercised when working with the equipment. Although every practicable safety precaution has been incorporated in this equipment, the rules under 1-1.2 and 1-1.3 below must be strictly observed.

1-1.2 Keep Away From Live Circuits.

Operating personnel must at all times observe all safety regulations. Do not make adjustments inside equipment with voltage supply on. Under certain condition dangerous potentials may exist in circuits with power controls in the off position. To avoid casualties, always remove power and discharge ground circuits prior to touching them.

1-1.3 Don't Service or Adjust Alone.

Under no circumstances should any person reach within the enclosure for the purpose of servicing or adjusting the equipment without the presence or immediate assistance of another person capable of rendering aid.

1-2. GENERAL DESCRIPTION

1-2.1 The Manual Control Assembly described herein is an electrical control assembly for the regulation of motor speeds of the Acoustic Device A Mark 6(b) NM Mod 1 and consists of a motor controller, an inverse current blocking rectifier assembly, and a dynamic braking resistor assembly, each contained in a separate enclosure. The motor controller is housed in a metal cabinet. The operator's controls are mounted on a hinged panel. The equipment panel for the controller circuit components is mounted on the inside of the cabinet. A hinged watertight hatch covers the control panel. The control panel swings open on two hinges to allow access to the equipment panel. The controller is connected to other units in the acoustic system by watertight cables which pass through stuffing tubes in the bottom of the cabinet. Submersible connectors and stuffing tubes are used in connecting the cable between the controller and the acoustic device.

1-2.2 The motor controller is equipped with an ammeter to measure armature current in the acoustic device and a tachometer indicator to

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register the revolutions per minute of the drive shaft of the A Mark 6(b) acoustic device. The control assembly circuit includes an inverse-current blocking circuit to protect the generators supplying power to the acoustic device, and a dynamic braking circuit to prevent overspeeding the acoustic device. The acoustic device motor is protected by an overload relay, and the 115-volt input to the controller passes through a circuit breaker. A flood switch in the housing of the acoustic device operates a relay in the controller which opens the power input circuit when the housing is flooded. The power for the controller is supplied by the ship's generators. The main controller elements are: an assembly of relays and a rheostat for field-circuit control of the acoustic motor, an ammeter, a tachometer indicator, inverse current and a dynamic braking circuit, and the necessary manual switches and indicator lamps.

1-3. DETAILED DESCRIPTION

The control assembly consists of the functioning components described in the following subparagraphs. See figures 1-1 and 1-2 for location of these parts.

1-3.1 Rheostat (figure 1-1)

The rheostat is a manually operated 50-step, 0 to 260-ohm plate type and is used to control the field circuit of the motor. The field current with all external resistance removed is approximately 2.2 amperes. The rheostat is located on the control panel.

1-3.2 Ammeter (figure 1-1)

The ammeter is a 0 to 100-ampere meter with a 50-millivolt external shunt. The ammeter registers the amount of current flowing in the armature circuit of the motor. The ammeter is located on the control panel.

1-3.3 Tachometer Indicator (figure 1-1)

The tachometer indicator is a 0 to 1-milliampere milliammeter calibrated for use as a speed indicator. The scale of the meter is calibrated to register in cycles per second (c.p.s.) from 0 to 40 and revolutions per minute (r.p.m.) from 0 to 2400. The tachometer indicator is located on the control panel.

1-3.4 Circuit Breaker (figure 1-1)

The circuit breaker is a 3-pole, 50-ampere, 120-volt d.c. air circuit breaker overload relay with one coil and a set of normally-closed contacts. The relay opens the circuit to the accelerating relays when a current overload occurs and is closed and reset manually.

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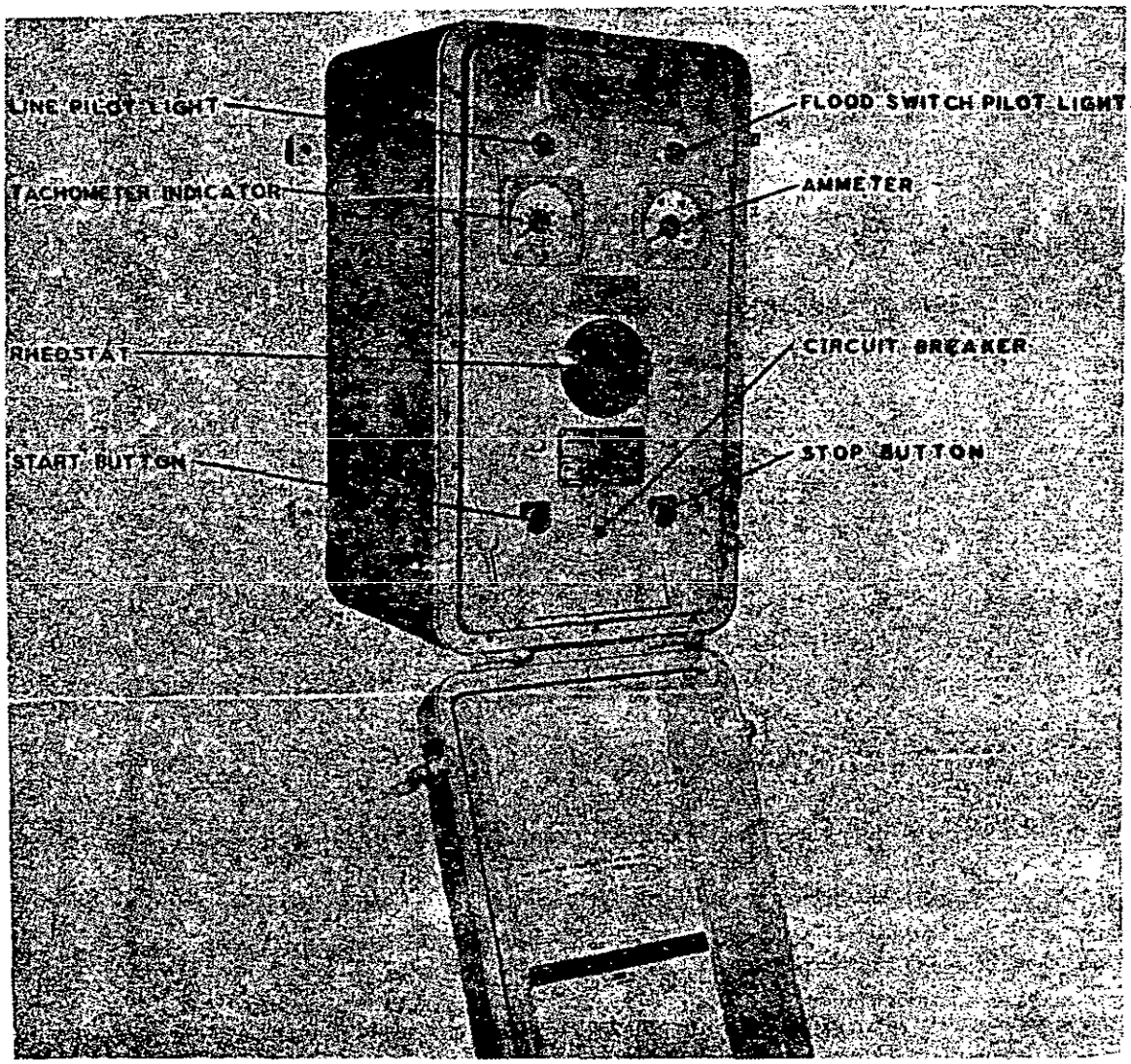


Figure 1-1. Control Panel

1-3.5 Pushbutton (Start-Stop) Station (figure 1-1)

The pushbutton station is a two-position switch with two pushbuttons, one used for starting and the other for stopping the acoustic device motor. The pushbuttons are located on the control panel.

1-3.6 Pilot Light (White) (figure 1-1)

The white pilot light indicates connection between the controller and the source of d.c. power. The pilot light is mounted on the control panel.

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1-3.7 Pilot Light (Red) (figure 1-1)

The red pilot light indicates closing of the flood switch. The pilot light is mounted on the control panel.

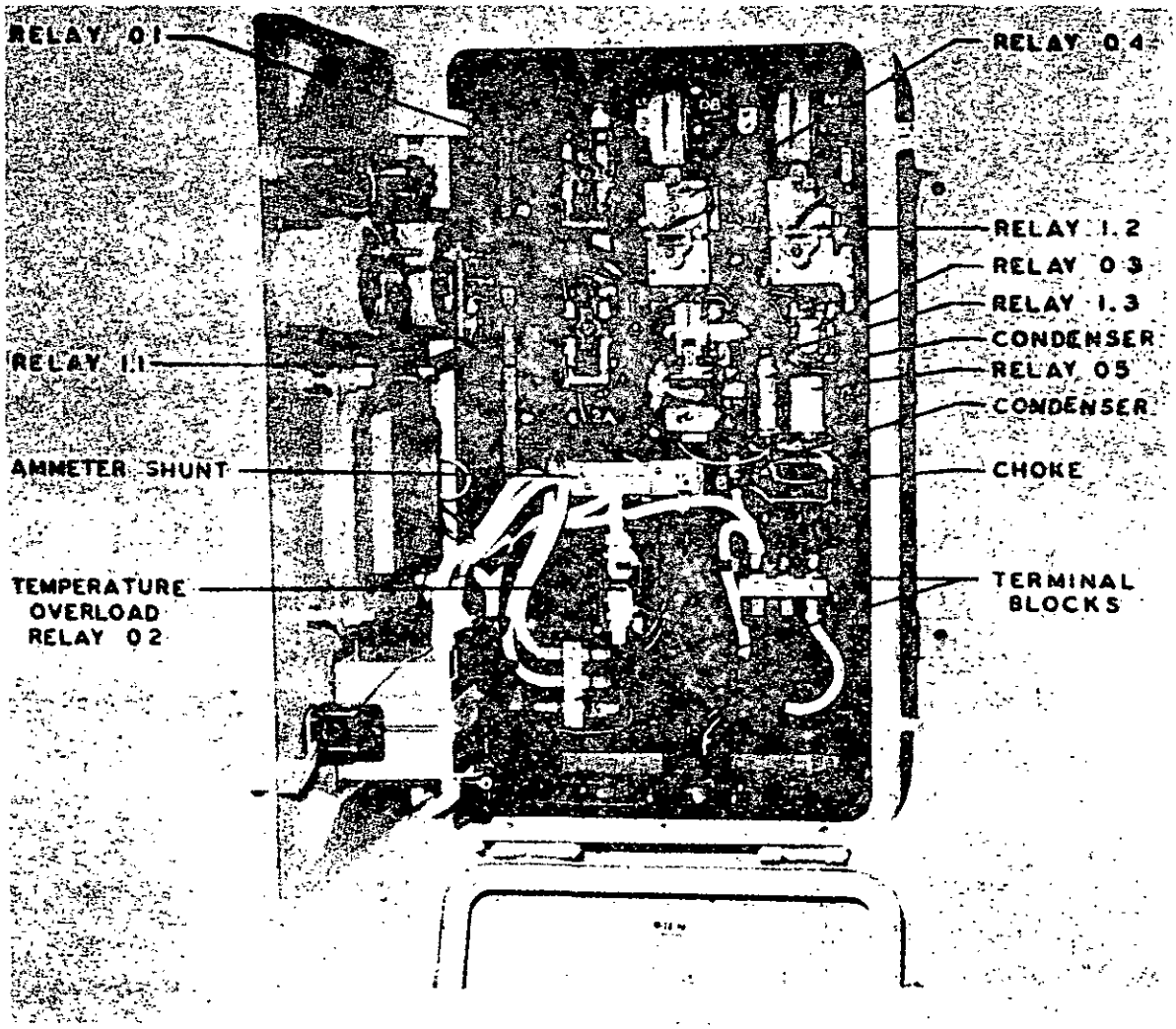


Figure 1-2. Inner Panel

1-3.8 Relays (figure 1-2)

(a) Relays 01 and 01.1 are time-accelerating relays, each having one coil, one set of normally-open contacts, and one set of normally-closed contacts. These relays function only in the starting of the acoustic device motor. These relays are mounted on the interior controller panel.

(b) Relay 01.2 the line contractor, has two coils, two sets of normally-open contacts, and one set of normally-closed contacts. This relay completes the circuit to the armature of the motor of the

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acoustic device, and is used to complete the starting of the motor. This relay is mounted on the interior controller panel.

(c) Relay 01.3, the full field relay, has one coil and one set of normally-open contacts. This relay shorts out the field rheostat while the motor is starting. This relay is mounted on the inner controller panel.

(d) Relay 03, the flood switch relay, has one coil, and one set of normally-open contacts, and one set of normally-closed contacts. This relay is operated by the flood switch in the acoustic device. When energized, it closes the circuit to the red pilot light and opens the circuit to the 01.2 relay coil. This relay is located on the inner panel of the controller.

(e) Relay 04, the dynamic brake relay, is the same type relay as 01.2. This relay closes the circuit to the Resistor Assembly. The relay is located on the inner panel of the controller.

(f) Relay 05, the detector circuit relay, is a hermetically-sealed relay with one set of normally-open contacts. This relay is operated by the inverse current flow from the motor armature. This relay is located on the inner panel of the controller.

1-3.9 Rectifiers

(a) Rectifier R1 is a type 1N93 diffused germanium rectifier. This rectifier prevents the flow of current from the positive side of the line to the 05 relay coil. This rectifier is mounted on the inner panel of the controller in the same enclosure as relay 05.

(b) Rectifiers R2, the inverse current rectifiers, are two selenium plate rectifiers. These rectifiers prevent the flow of inverse current from the armature from entering the positive side of the line through the 01.2 contacts and the ammeter. These rectifiers are located in the inverse current blocking rectifier assembly.

1-3.10 Brake Resistors

The brake resistors are three strip-type resistors, each rated at 1.74 ohms at 22 amperes. These resistors dissipate the inverse current from the motor. The resistors are located in the Resistor Assembly, A Mark 6(b) NM, Mod 1.

1-3.11 Choke

The choke has a rating of 150 henries inductance, 10 ma d.c. current, 850 ohms resistance. It is used to prevent the flow of alternating current from tachometer indicator, thereby preventing chattering of the flood switch relay. The choke is mounted on the inner panel of the controller.

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Part 2. Theory of Operation

2-1. GENERAL

This part describes the electrical theory of operation of the manual control assembly used to operate the Acoustic Device A Mark 6(b) NM Mod 1. Figure 2-1 is a block diagram and figure 2-2 is a schematic diagram of the complete control system, showing the interconnections between the control assembly, the automatic controller, and the acoustic device motor. A sealed-in schematic diagram and an operating instruction sheet are also included in the cabinet of the controller. The automatic controller is directly-wired into the circuit between the control assembly and the acoustic device, making it possible to switch over from manual control to automatic control of the motor. The electrical functioning of the automatic controller is described in this manual only to the extent that it directly affects the operation of the control assembly. A full description of the automatic controller circuit can be found in chapter 13, section 13-202 of NAVSHIPS 250-530 covering operation of the automatic controller. The electrical input to the manual controller is 115 volts d.c.

2-2. ELECTRICAL FUNCTIONING

2-2.1 General

The control assembly is used for a semiautomatic regulation of the speed of the drive motor of the acoustic device. This is accomplished by varying the resistance in the field circuit of the motor. Adding resistance in the field winding of the motor reduces the field strength, decreasing the counter electromotive force in the armature circuit, thereby permitting more current to flow in the armature. This increased armature current speeds up the motor. When the motor is started, resistance is inserted in the armature circuit by means of a set of relays. Paragraph 2-2.2 describes the operation of the control assembly.

2-2.2 Operation of the Control Assembly

(a) When the circuit breaker is closed, using the toggle switch, current flows from the 115-volt d.c. positive line (+) through the white pilot light (W) to the 115-volt d.c. negative line (-).

(b) When the start button is pushed, the following sequence occurs:

First accelerator contactor 01 opens its high current contacts and closes its pilot contacts which energizes the second accelerator contactor 01.1. The high current contacts of 01.1 open and the pilot contacts of 01.1 close which energizes main line contactor 01.2. This action closes the line contacts 01.2 thus connecting the positive side of the line to the acoustic device through the starting resistances. Pilot contacts of 01.2 close the hold circuit of the line contactor and open the first accelerator coil circuit. After a delay of approximately 2 seconds, contactor 01 closes its high current contacts, cutting out 67 percent of the armature

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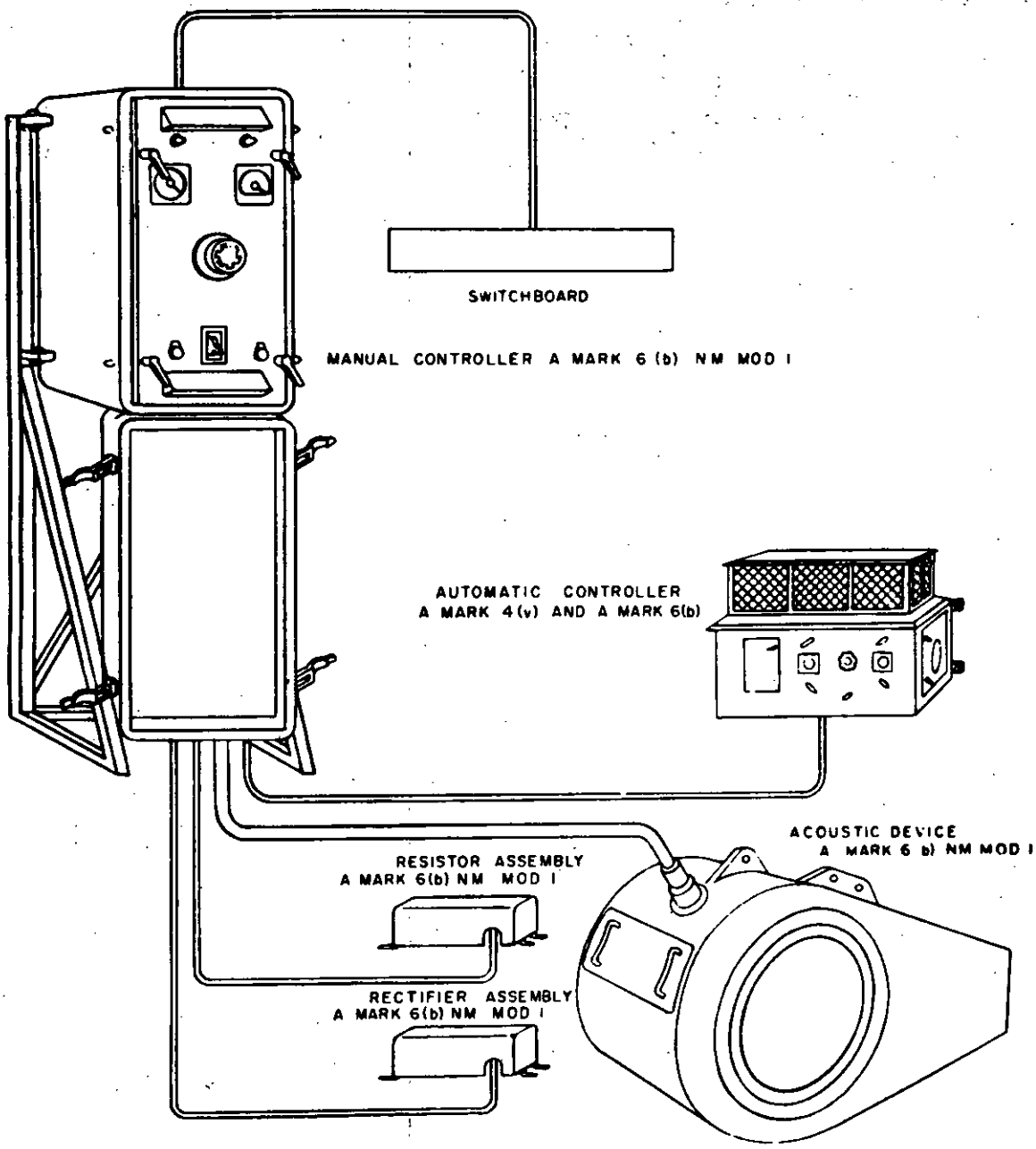


Figure 2-1. Block Diagram

series resistance together with opening pilot contacts which de-energize the contactor 01.1. After a delay of approximately 2 seconds, contactor 01.1 closes, cutting out the remaining series resistance together with opening pilot contacts which de-energize the contactor. During the acceleration period, full field contacts 01.3 were closed by the 01.3 coils which are voltage actuated by the drop across the armature series resistors.

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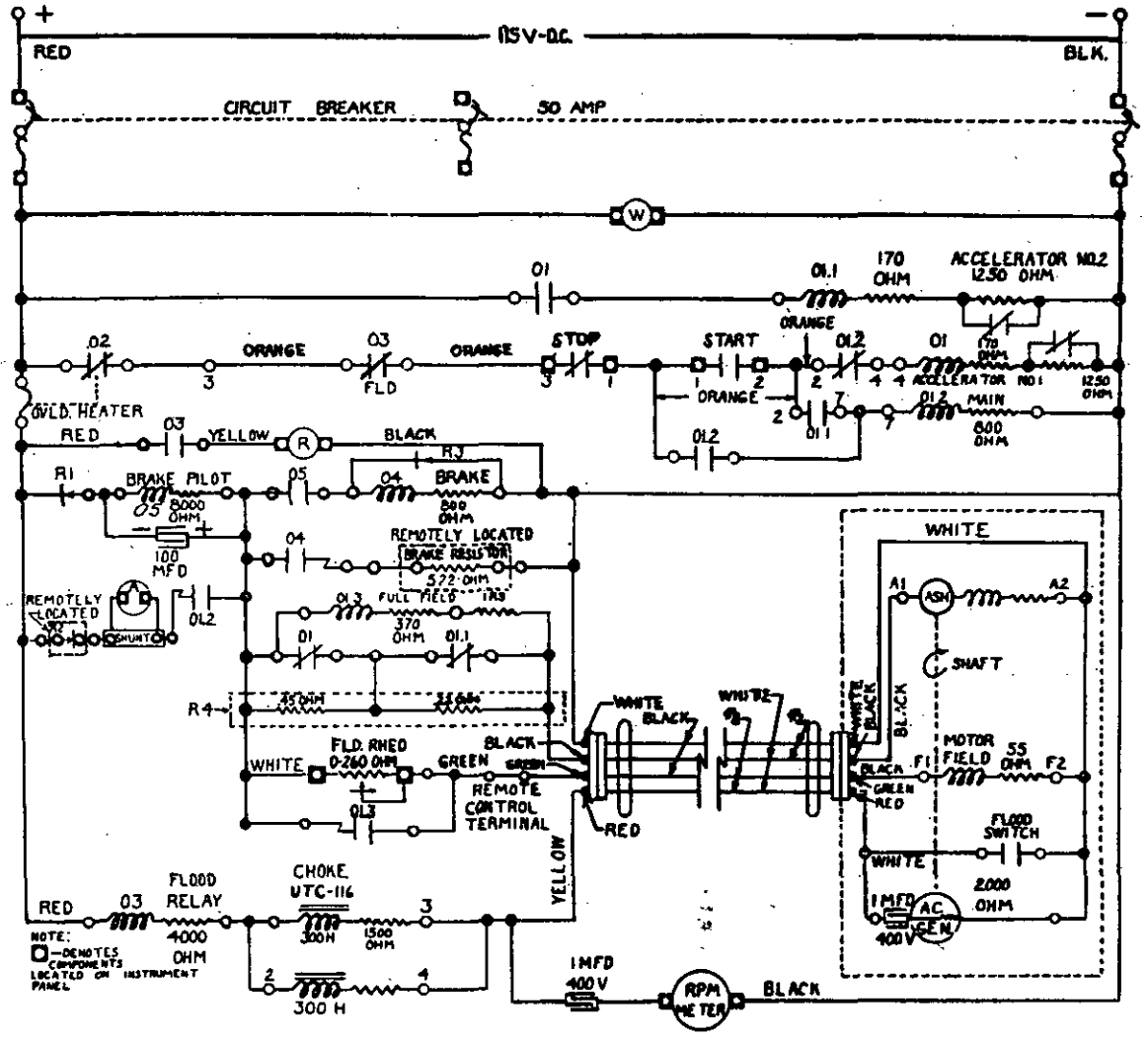


Figure 2-2. Schematic Diagram

(c) The 1250 ohm resistors in series with accelerator coils 01 and 01.1 are provided to reduce heating in the coils. Normally closed auxiliary contacts short out these resistors to allow the full initial current which is required to pick up the accelerator contacts. After the initial rush of current the auxiliary contacts open placing the 1250 ohm resistors in series with coils 01 and 02, thus reducing the current to a value less than that required to pick up the contacts but sufficient to hold them.

(d) The inverse current blocking and dynamic braking circuit operates as follows:

Rectifier R2 passes current so that the motor may operate, but whenever the field is strengthened and the armature current tends to reverse it is instantly blocked. During this time, a voltage of such polarity is developed across R2 as to cause current flow through rectifier

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R1, hence, charging the 100 MFD capacitor and, if greater than 16 volts, operating brake pilot relay 05. The closing of contacts 05 energizes coil 04, closing contacts 04 which shunt the armature with 5.22 ohms resistance. This brake resistor slows the motor to base speed within 5 seconds. Shortly after the voltage across R2 has disappeared, the 100 MFD capacitor discharges through coil 05 and contacts 05 open releasing contacts 04, hence removing the brake. Rectifier R3 is used to suppress arcing at contacts 05 due to the inductance of coil 04.

2-2.3 Stopping the Controller and Acoustic Device

When the stop switch is pressed contacts 3 and 1 are opened, the operation of the control assembly and the acoustic device are as follows:

The current supply to the coil of relay 01.2 is interrupted, causing its contacts to return to their normal position, and the current supply to the armature is interrupted, causing the motor to slow down gradually.

2-2.4 Flooding Protection

Water entering the acoustic device housing operates the flood switch and completes an operating path for the 03 flood relay. Current flows from the 115-volt d.c. positive line as follows:

(a) Through the 03 flood relay coil, through UTC 116 choke 1-3 to the flood lead terminal on the manual controller, through the cable and connectors to terminal SW on the acoustic device, the flood switch, to terminal A2F2 through the cable and connectors to the line-terminal of the manual controller.

(b) The flood relay operates and closes its normally-open 03 contacts and opens its normally-closed 03 contacts, stopping the current supply to the 01.2 relay coil, and opening the motor armature circuit.

(c) Current flows from the 115-volt d.c. positive line (+) through the 03 contacts through the red pilot light (R) to the 115-volt d.c. negative line (-).

Part 3. Operating Instructions

3-1. GENERAL

The operating instructions assume that after installation, the equipment has been checked for proper connections and tested with the power on and that the unit is in proper operating condition. Before operating this equipment, the user should become thoroughly familiar with the theory of operation and the purposes of the system in which it functions, as described in Part 2, Theory of Operation, and in this part. This part presents each step in the operating sequence employed for use with the system employing

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A Mark 6(b) type equipment. These steps should be followed in the sequence given in order to obtain the desired results and to avoid possible malfunctioning or damage to the equipment. The operating instructions for associated units, the acoustic devices and the automatic controller, are included in the sections dealing with those units. Table 3-1 is an operational summary of all front-panel controls.

TABLE 3-1. Operational Summary of Front Panel Controls (see Figure 1-2)

Panel Control	Function
Start-Stop Pushbutton Station	A two-position pushbutton station, with a separate button for starting and one for stopping the motor.
Pilot Light (White)	A white lamp used to indicate that power is connected to the controller.
Pilot Light (Red)	A red lamp used to indicate closure of the flood switch in the acoustic device housing, and hence presence of sea water in the housing.
Ammeter	A d.c. ammeter reading from 0 to 100 amperes for measuring current flow in the armature circuit of the acoustic device motor.
Tachometer Indicator	A d.c. milliammeter calibrated to convert amperage readings into revolutions per minute, and used to measure the r.p.m. of the acoustic device.
Circuit Breaker Switch	A two-position toggle switch operated by the circuit breaker to open the operating circuit of the controller and acoustic device.
Field Rheostat	A 0 to 260-ohm rheostat for manually adjusting the speed of the acoustic device motor by varying the resistance in the field circuit.

3-2. OPERATING PROCEDURE

The control assembly is wired and adjusted by the manufacturer before shipment and is ready for operation after interconnection with the acoustic system has been performed. Make the following settings and adjustments to operate the A Mark 6(b) type Acoustic Devices.

3-2.1 Turn the field rheostat fully counterclockwise (all resistance out).

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3-2.2 With the S1 left hand contacts switch on the automatic controller in the "OUT" position, and the circuit breaker switch closed, push the "Start" button.

3-2.3 The motor will start and come up to normal speed within 3 to 5 seconds. Starting current indicated on the ammeter will be about 60 to 70 amperes, and will diminish to the current normal for the running speed.

3-2.4 Observe the tachometer readings after the motor has reached its normal operating speed. The reading should be approximately 700 r.p.m.

3-2.5 If the left hand face plate of the automatic controller is to be used for field control of the acoustic drive motor, the automatic controller circuit must be switched into the control system. This should be done according to the procedure described in the sections on the automatic controller.

3-2.6 To stop the acoustic device, press the "Stop" button momentarily.

NOTE

a. If the starting current does not drop down from 60-70 amperes to normal within 5 seconds, press the stop button immediately; check for the trouble according to procedure for Corrective Maintenance Part III, Chapter 23, Section 23-224 before restarting the motor.

b. If a current overload occurs during operation, the overload relay will operate and break the motor circuit. Check for trouble according to procedure for Corrective Maintenance, Part III, Chapter 23, Section 23-224. Reset the overload relay by pushing the circuit breaker toggle switch on the control panel to the "ON" position.

3-3. TEST PROCEDURE

The A Mark 6(b) Type Manual Controller, the A Mark 6(b) Type Acoustic Device and the Automatic Controller comprise one acoustic sweep system. The units must be connected in accordance with figures 2-1 and 2-2, and tested immediately prior to each minesweeping operation on the minesweeper, or at dockside.

3-3.1 The acoustic device shall be made ready for operation in accordance with the appropriate section of NAVSHIPS 250-530, Chapter 13.

3-3.2 The system can be checked while the acoustic device is on deck, or while it is in water. If the device is to be on deck during the test, the air pressure must be released from the housing. If the device is to be in water, the housing must be pressurized to balance the external water pressure at the depth to which the device is to be submerged.

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APPENDIX II

This appendix is an example of the type of information required in the Technical Manual for maintenance of the manual controller.

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NOTE

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b. If a current overload occurs during operation, the overload relay will operate and break the motor circuit. Check for trouble according to procedure for Corrective Maintenance, Part III, Chapter 23, Section 23-224. Reset the overload relay by pushing the circuit breaker toggle switch on the control panel to the "ON" position.

3-3. TEST PROCEDURE

The Control Assembly, Manual A Mark 6(b) NM Mod 1 should be tested immediately prior to actual operation on a sweep vessel. The tests should be conducted with an assembled acoustic sweeping system ready for operation. As each of the units in an acoustic minesweeping system is dependent for its proper functioning on the other units, the test procedure applies to the system as a whole. To check for proper operation of the acoustic system, start the acoustic device motor and check the current readings. The current should rise to 60-70 amperes for 5 seconds and then drop to normal current drain during normal running. Check to see that the operating speed is within 680 to 725 r.p.m. If the current readings are too high, or speed is outside the specified range, check the system according to the procedure for Corrective Maintenance, Part III, Chapter 23, Section 23-224.

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OPERATION AND MAINTENANCE OF MINE COUNTERMEASURES EQUIPMENT

Part III. Maintenance of Mine Countermeasures Gear and Auxiliary Equipment

Chapter 23. Acoustic Minesweeping Gear

SECTION 23-224

MANUAL CONTROL ASSEMBLY - A MARK 6(b) NM MOD 1 (MANUFACTURED BY MECO, INC.)

NAVAL SHIP ENGINEERING CENTER
CENTER BUILDING
PRINCE GEROGE'S CENTER
HYATTSVILLE, MARYLAND 20782

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Part 1 - PREVENTIVE MAINTENANCE

1-1. SAFETY PRECAUTIONS

1-1.1 General. a. The attention of officers and operating personnel is directed to Chapter 18 of the United States Navy Safety Precautions (OPNAV 34P1) or superseding instructions on the subject of safety precautions to be observed.

b. This equipment employs voltages which are dangerous and may be fatal if contacted by operating personnel. Extreme caution should be exercised when working with the equipment. Although every practicable safety precaution has been incorporated in this equipment, the following rules must be strictly observed:

1-1.2 Keep Away from Live Circuits. Operating personnel must at all times observe all safety regulations. Do not make adjustments inside equipment with voltage supply on. Under certain conditions dangerous potentials may exist in circuits with power controls in the off position. To avoid casualties, always remove power and discharge ground circuits prior to touching them.

1-1.3 Don't Service or Adjust Alone. a. Under no circumstances should any person reach within or enter the enclosure for the purpose of servicing or adjusting the equipment without the presence or immediate assistance of another person capable of rendering aid.

b. When any electrical equipment is to be worked on, it shall be disconnected from the source of supply by opening main or branch supply switches, circuit breakers, or cutouts so as to eliminate the possibility of current flowing to the equipment. Switches, circuit breakers, or cutouts opened for this purpose must be secured in the open position and must have tags attached, and only the individual placing the tag shall remove it and re-energize the circuit.

c. After the switches are secured, they should be tagged with a "Hold" card bearing the following: "This circuit was ordered open for repairs and shall not be closed except on direct orders of the undersigned."

d. The "Hold" card must be signed and dated, usually by the person directly in charge of the work. If the card cannot be seen clearly, a red light should be used to illuminate it.

e. When the work has been completed, the tag should be removed by the same person. No person shall remove another person's tag. If the switch has been locked, it should be unlocked and the circuit restored by the person retaining the key.

f. When more than one group is engaged in the work, a signed and dated tag for each group shall be placed on the supply switch. The individual who tagged this switch for his group should remove only his own group's tag when the work has been completed and it has been determined that all persons are in the clear.

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1-2. GENERAL PROCEDURE

A series of preventive maintenance operations should be carried out periodically to insure maximum operating efficiency of the control assembly. The procedure consists of inspection of the controls and functioning parts and simple replacement of worn or damaged parts. The method of replacement of worn or damaged parts is described in Part 2, Corrective Maintenance. Be sure that the power supply to the controller is disconnected and the circuit breaker switch is turned to the "OFF" position before any attempt to inspect or to make adjustments to any internal parts.

1-3. CHECKING PROCEDURES

Check the control panel (figure 1-1) for broken lights, damaged, or worn controls. Check the rheostat knob for slippage and tighten set screws if loose. Replace parts where necessary.

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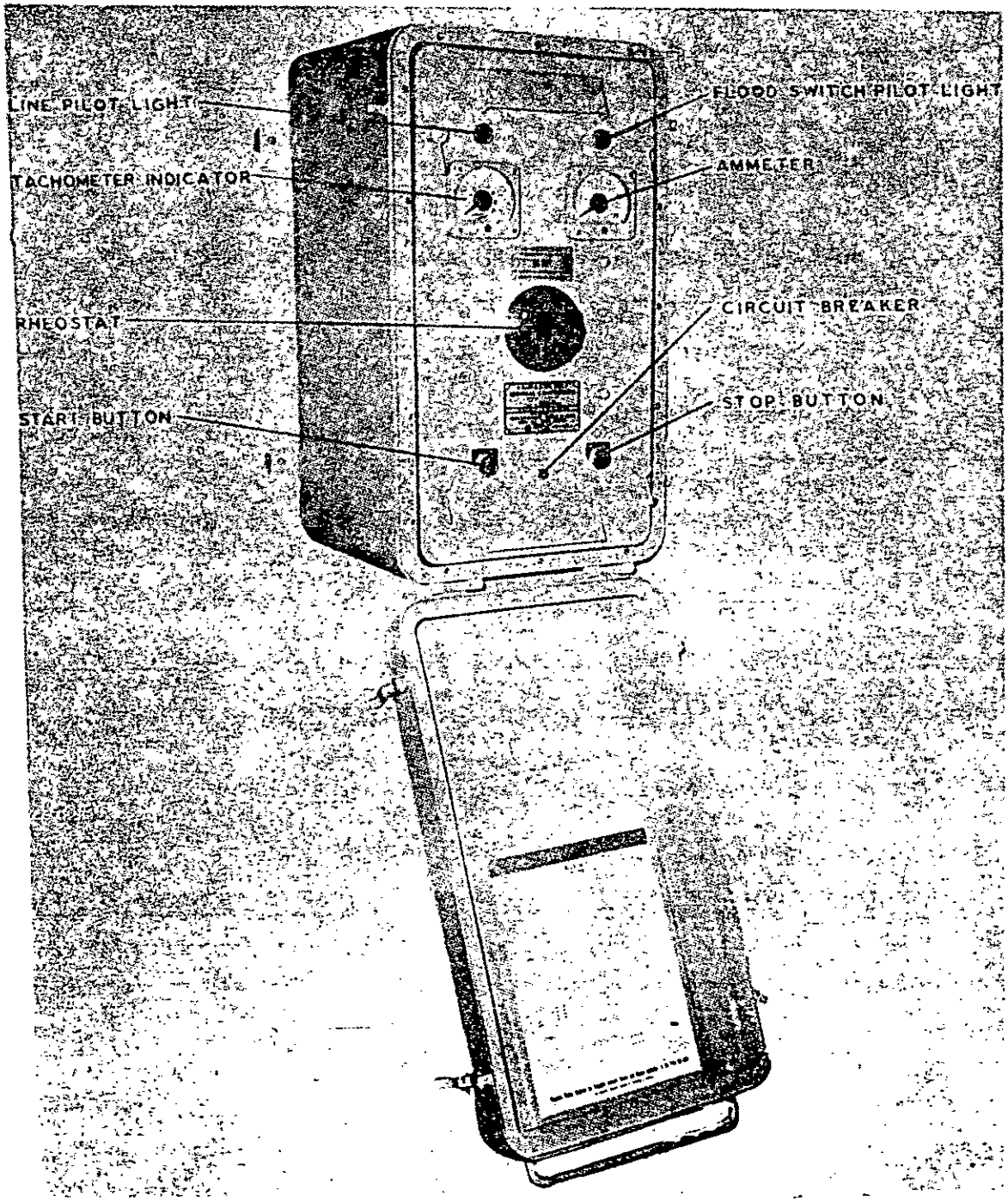


Figure 1-1. Control Panel

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Part 2 - CORRECTIVE MAINTENANCE

2-1. TROUBLE SHOOTING

2-1.1 Trouble Location. Failure of any of the components of the control assembly will usually register as malfunctioning or complete interruption of service of the acoustic minesweeping system. To isolate troubles in the assembly, first locate specific operating symptoms. Use the trouble shooting table to localize and correct trouble occurring within the acoustic device. When the source of malfunctioning has been determined, repair the controller as indicated in the table.

2-1.2 Trouble Correction. a. Malfunctioning parts should be located in accordance with table 2-1 and should be repaired or replaced as indicated therein.

TABLE 2-1

TROUBLE SHOOTING TABLE

Symptom	Probable Cause	Corrective Measure
Push button depressed to "ON" position, acoustic device does not run, pilot light off.	1. Open circuit between 115-volt source and controller. 2. Defective push button contacts.	1. Locate opening in circuit and close. 2. Clean contacts. If this does not remedy difficulty replace push button.
Push button in "ON" position, acoustic device does not run, pilot light on.	1. 90-ampere fuses blown. 2. Loose connections at control terminal block. 3. Defective push button. 4. Defective field rheostat.	1. Locate cause of fuse blowing. Correct and then replace fuse. 2. Tighten connections. 3. Replace push button. 4. Replace rheostat.
Excessively high current reading on ammeter during normal running operation.	1. Low pressure in acoustic device main housing. 2. Defective field rheostat. 3. Defective ammeter.	1. Recover acoustic device and check pressure to assure accordance with operating instructions. 2. Replace rheostat. 3. Replace ammeter.
Irregular current reading and motor operation as field rheostat is rotated. This condition may end in blown fuses.	1. Defective field rheostat.	1. Replace rheostat.
Current reading on ammeter too low. Tachometer indicates low motor speed. This condition might result in a burned out resistor.	1. Defective operation of relays in controller circuit. If 01 is defective, it may keep the .45-ohm resistor in the armature circuit at all times, slowing down the motor. If 01.1 is defective, it may keep the .22-ohm resistor in the armature circuit at all times, slowing down the motor.	1. Clean relay contacts and tighten all connections. If this does not correct difficulty, repair or replace relays as necessary. Spare contacts and springs are furnished with the controller as on-board repair parts.

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TABLE 2-1 (Cont)

Symptom	Probable Cause	Corrective Measure
	A combination of the above defects in the contact can keep both .45 and .22-ohm resistors in the armature circuit, slowing down the motor somewhat.	
Interruption of operation. Equipment found in satisfactory working condition.	1. Defective overload relay contacts. 2. Defective flood relay contacts. 3. A random break in the lines inside or outside controller.	1. Clean or replace contacts. 2. Clean or replace contacts. 3. Locate break and repair.
Interruption of operation by overload relay. Equipment found in satisfactory working condition.	1. Low supply voltage.	1. Find cause of low voltage and correct.

b. Open control panel as indicated in figure 2-1 and examine inside components for signs of overheating, breaks, or short circuits between or within parts. Check wiring for frayed insulation or broken connections. Check all relay contact surfaces for cleanliness and uniformity of mating surfaces. Check the contact impact springs for even tension and renewal of defective springs. Check all protective devices for proper operation. When checking, be sure to note that all relays operate in their proper sequence. Lubricate sparingly, especially at the hinge pins of contactors, levers and armature, taking particular care to wipe off all excess oil. Avoid use of excessive oil at all times. Excessive oil is harmful because the oil creeps and collects dirt, thus forming current leakage paths which eventually result in breakdown of the insulating materials.

c. Checking of the rheostat is essential. The rheostat should be kept dry and as free from dust as possible. All exposed molded insulation surfaces should be wiped off frequently. If broken or cracked, they should be promptly renewed. If the rheostat should get wet, it should be wiped carefully and dried before being put back into service. Whenever convenient, the rheostat should be dried in a warm compartment. The rheostat also may be warmed by allowing a low current to pass through the windings until all moisture has been expelled. Start with about one-fourth of the rated current and, as the rheostat warms up (if no trouble develops), gradually increase the amperage until full rated current is allowed to flow. This condition should continue until the insulation readings are one megohm or higher. Use a painter's duster to remove dust and dirt. Blow dust and dirt from inaccessible parts with dry compressed air.

d. Check wiring outside of controller for broken cable connections or frayed or shorted leads. Repair or replace cable and connectors where necessary.

e. When the control panel is closed, precaution should be taken to prevent pinching any of the electrical leads between the panel and the

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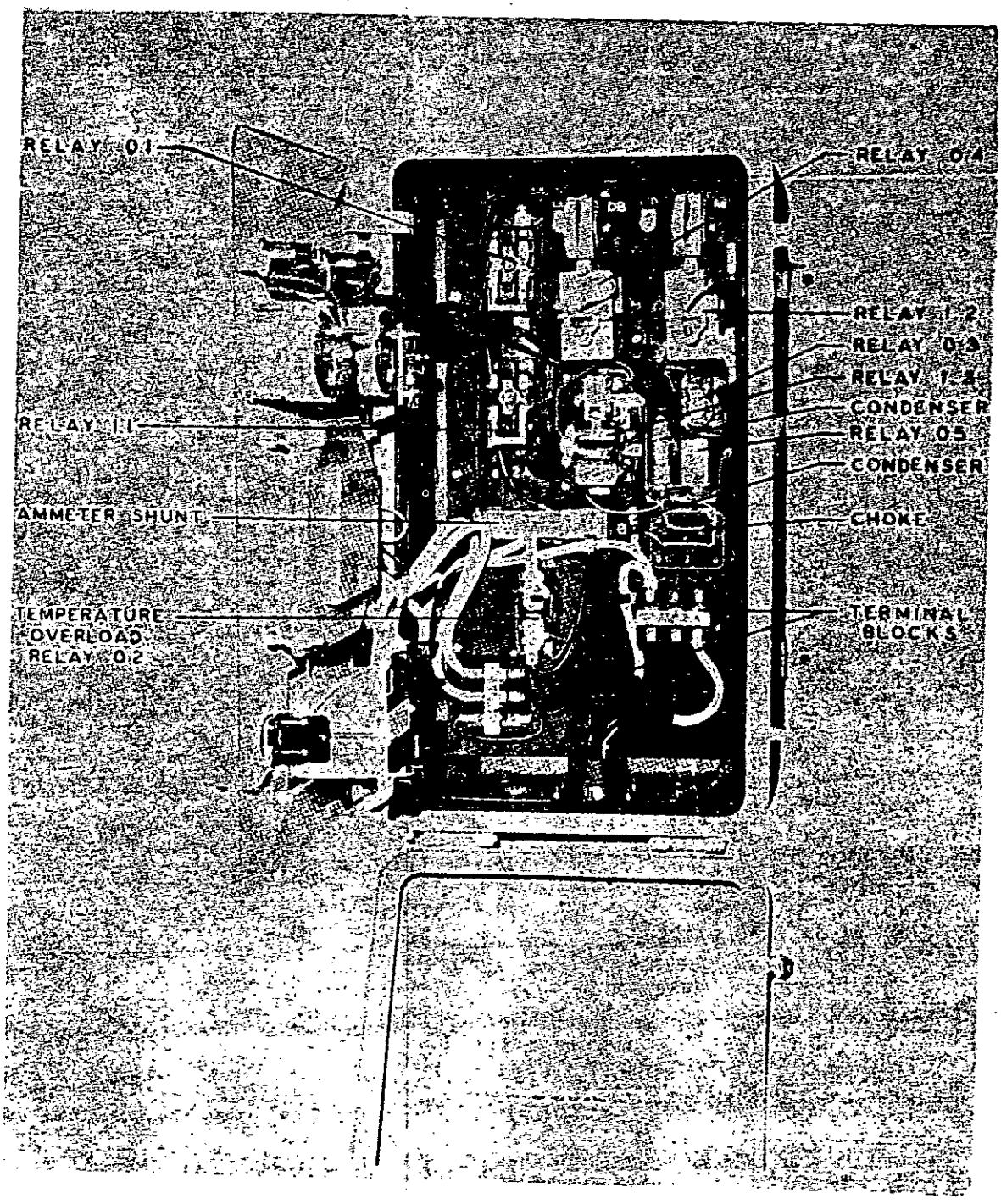


Figure 2-1. Inner Panel

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cabinet. Pinching leads may result in a ground or short circuit which will be followed by unsatisfactory operation of the controller.

2-2. REMOVAL, INSPECTION AND REPLACEMENT OF COMPONENTS

2-2.1 Removal. a. The operating controls and lamps on the control panel are accessible for removal from the rear of the panel. Open the control panel first by removing the wing nuts and loosening the fasteners. The cover can then be swung outward on its hinges. To remove controls from the panel, disconnect leads and remove mounting screws or bolts.

b. The various functioning components in the cabinet of the manual controller are accessible for servicing when the control panel is opened. To remove any of the electrical parts on the equipment panel, simply disconnect the leads from the parts and remove the mounting screws and/or nuts.

2-2.2 Inspection. Equipment and control parts may be inspected while in position on the panels. Visually inspect all wiring for evidence of overheating, short circuits, broken connections or leads, and frayed insulation. Check parts for evidence of overheating, mechanical damage to movable contacts, etc., and evidence of frozen contacts and short circuits. Replace all parts which cannot be repaired by reconnecting, reinsulating, or cleaning and adjusting. Check the controller for proper functioning according to the procedure described in Part II, Chapter 13, Section 13-224, Article 3-3, before installing covers.

2-2.3 Replacement. After replacing all electrical parts on the control panel, be sure that the control panel is closed and latched before the controller is put into service.

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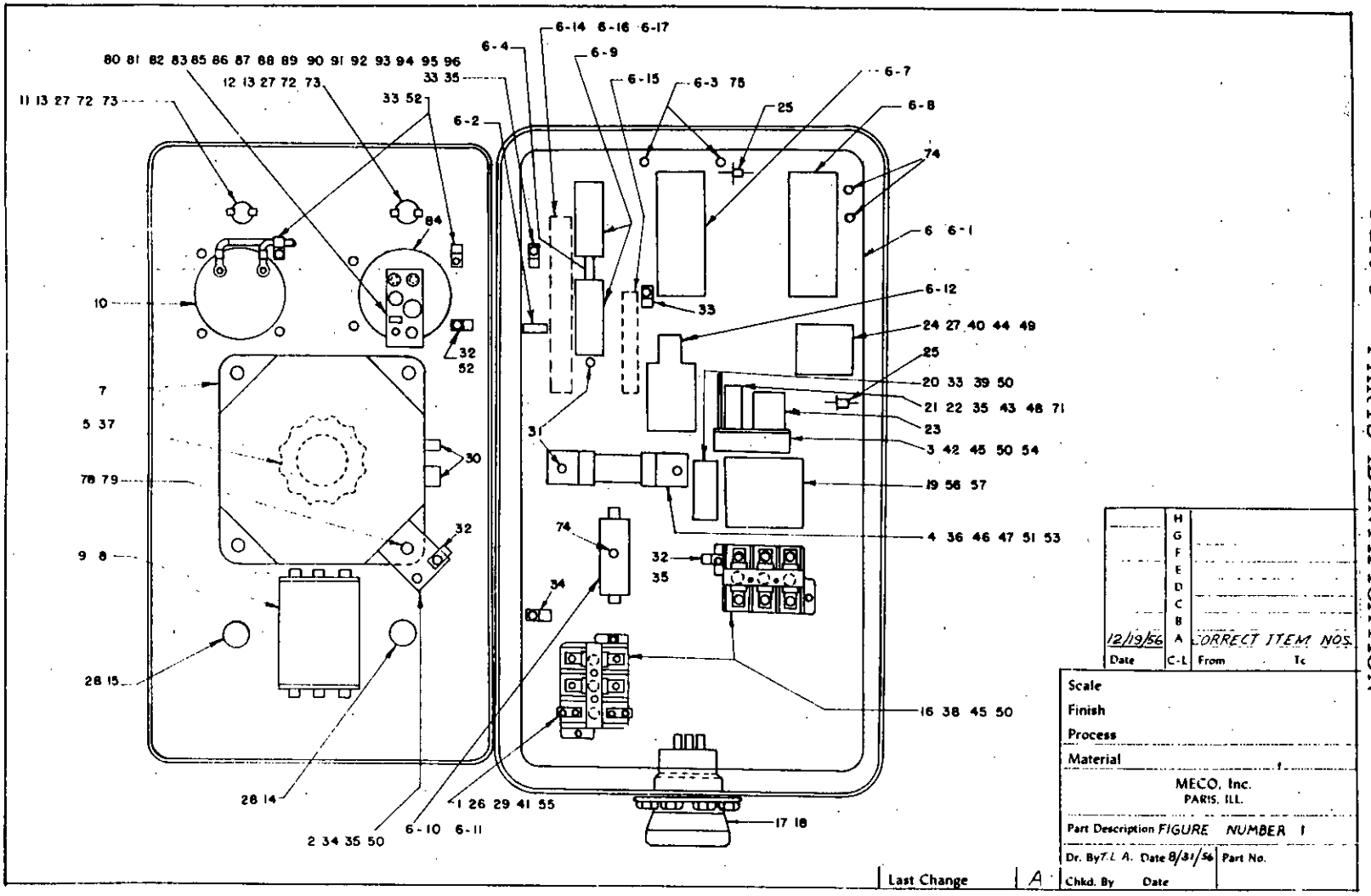


Figure 3-1. Manual Controller, A Mark 6(b) NM MOD 1, Electrical Components

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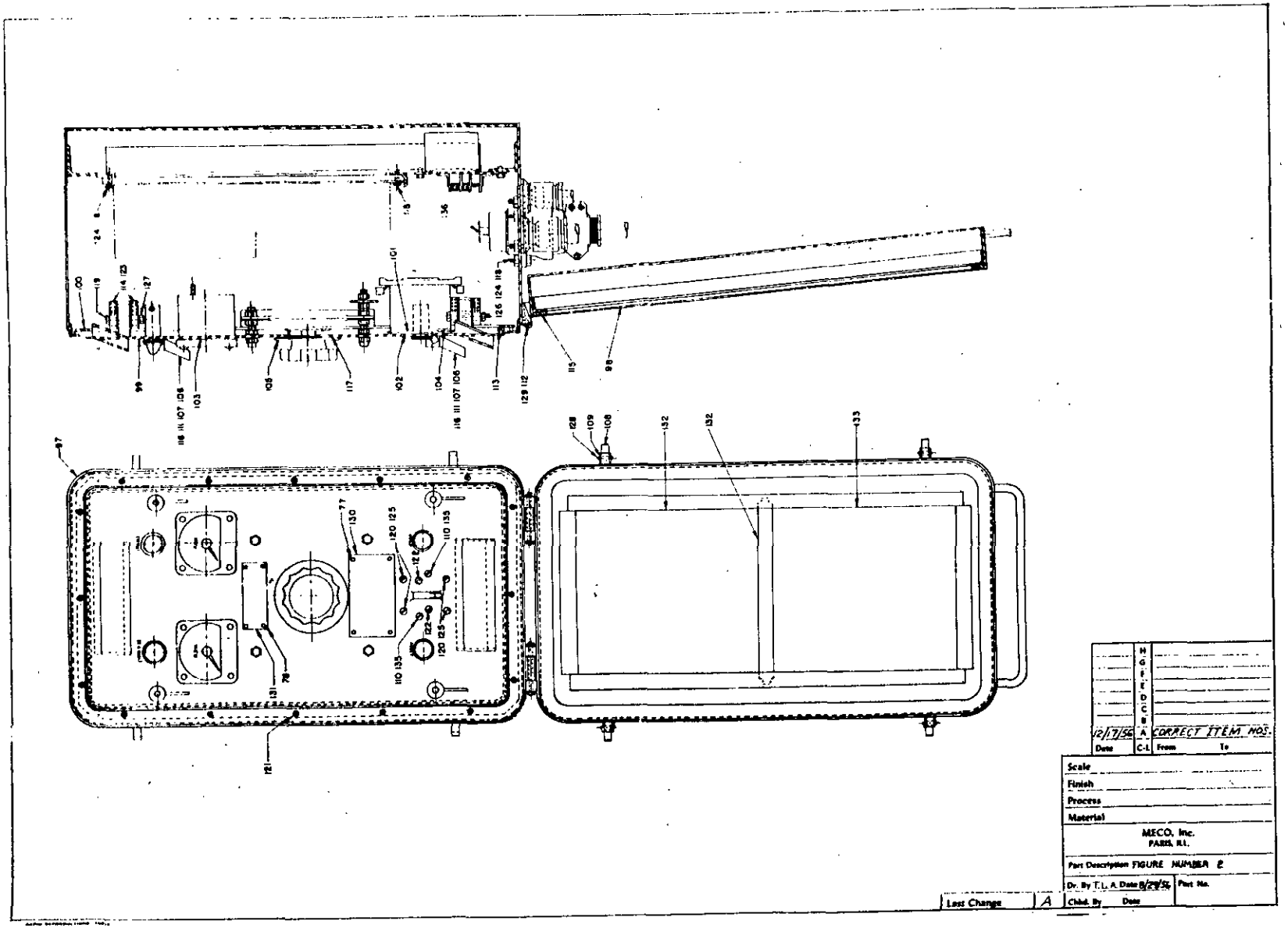


Figure 3-2. Manual Controller, A Mark 6(b) NM MOD 1, General Assembly

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TABLE 3-1

PARTS LIST, MANUAL CONTROLLER, A MARK 6(b) NM MOD 1

Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
3-1	1	Insert, Terminal Block	1 ea.	3223563-1	MALH	3223563-1		
	2	Bracket, Wire Clip	1 ea.	3223563-2	MALH	3223563-2		
	3	Bracket, Relay	1 ea.	3223563-3	MALH	3223563-3		
	4	Block, Shunt Mounting	2 ea.	3223563-4	MALH			
	5	Knob, Rheostat	1 ea.	3223563-5	(Reworked by Meco)	P9155374-1		
	6	Controller, Magnetic 115V, D.C. 7½ HP	1 ea.	3223563-6	GAAI		Control starting & operation of motor (Consists of Pc.Nos. 6-1 to 6-17)	
	6-1	Base, A19F1B Comp. 29 x 16 x ½	1 ea.	3223566-26	GAAI	182D232		
	6-2	Panel Nameplate	1 ea.		GAAI	NP69307C	Stamped per S.O.	
	6-3	Conn. Block	2 ea.		GAAI	L-6922211 G18		
	6-4	Conn. Strap (1A-2A)	1 ea.		GAAI	V-9098022P3		
	6-7	Contactor (DB)	1 ea.		GAAI	IC2800-1129A1	Coil No. 22D11G3	
	6-8	Contactor (M)	1 ea.		GAAI	IC2800-1129AM1	Coil No. 22D11G3	
	6-9	Accel. Contactor (1A,2A)	2 ea.		GAAI	IC5481-33F302		
	6-10	Relay (OL)	1 ea.		GAAI	CR2824-42B		
	6-11	Heater (OL)	1 ea.		GAAI	81D591		
	6-12	Relay (FF-FA)	1 ea.		GAAI	IC2820-1748B13		
	6-14	Start Resistor (SR)	1 ea.		GAAI	IC9033C3B67AC	Mounted on Back of Panel	
	6-15	Resistor (1RS)	1 ea		GAAI	9756510E39F	Mounted on Back of Panel	
	6-16	Support (SR)	1 ea.		GAAI	288A997 P1	Mounted on Back of Panel	
	6-17	Support (SR)	1 ea.		GAAI	288A997 P2	Mounted on Back of Panel	
	7	Rheostat, Field	1 ea.	3223563-7	GAAI	EB-101AA-62	Manual controls field current	
	8	Circuit Breaker 50 AMP	1 ea.	3223563-8	TRE	K173-A50	Protect controller and motor against high current	
	9	Sub Base, Line and Load Stabs for Back Connection	1 ea.	3223563-9	TRE	K250-2B		
	10	Ammeter, 50MV(Comp. with 4 ft. leads and shunt)	1 ea.	3223563-10	GEMI	8DB14A3AA1	Indicates amount of field current	
	11	Pilot Light, Red	1 ea.	3223563-11	DLC	41101-111		
	12	Pilot Light, White	1 ea.	3223563-12	DLC	41101-115		

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Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
3-1	13	Lamp, Incandescent, Cand. Screw	2 ea.	3223563-13	GLEC	6S6 (115V)		G-6240-797-8600
	14	Push Button (Oil Tight) Bull. No. 800T - Green	1 ea.	3223563-14	AB	2-5749		
	15	Push Button, Red (Oil Tight) Bull. No. 800T	1 ea.	3223563-15	AB	2-5728		
	16	Terminal Block, 3 Pole	2 ea.	3223563-16	SQDC	CL.9080-Ty.U-3		
	17	Connector, Receptacle	1 ea.	3223563-18	CS	6728		
	18	Connector, Plug	1 ea.	3223563-19	CS	6726		
	19	Choke 2-300M4 Coils & 2-1500 ohm resistors	1 ea.	3223563-20	UNT	HC-116		
	20	Condenser 1 MFD	1 ea.	3223563-21	ASTN	MQLF-4-1M		
	21	Condenser, Electrolitic 100 MFD 300 V., D.C.	1 ea.	3223563-22	AEV	AFH1-36	Charges and if charges too high operates brake pilot relay 05	
	22	Socket, Condenser "plug in"	1 ea.	3223563-23	CIN	2C5		
	23	Relay, hermetically sealed	1 ea.	3223563-24	SGI	5RJ	Operated to initiate braking action	
	24	Relay, 115V., D.C.	1 ea.	3223563-25	CUT	6001H615A518	Actuates by water across flood switch	
	25	Rectifier, diffused germanium	2 ea.	3223563-27	GEK	1N93	R1 Control operation of current flow to charge 100 μf condenser. Rc Suppress arcing at contacts of 05	
	26	"Lug-it" solderless lug (Electro-tinned)	2 ea.	3223563-29	TOM	35301		
	27	Lug, Plastic Grip ring tongue No. 6 Screw	15 ea.	3223563-30	ARP	34144		
	28	Lug, Plastic Grip ring tongue No. 8 Screw	8 ea.	3223563-31	ARP	34145		
	29	Lug, Plastic Grip ring tongue No. 10 Screw	11 ea.	3223563-32	ARP	34146		
	30	Lug, Plastic Grip ring tongue ¼	2 ea.	3223563-34	ARP	34150		
	31	Lug, Plastic Grip ring tongue ¼ No. 10 AWC	2 ea.	3223563-35	ARP	31127		
	32	"Ny-Grip" clamp 7/16	3 ea.	3223563-37	WECK	Type 6 NY Grip 7/16		

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Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
	33	"Ny-Grip" clamp 11/16	5 ea.	3223563-39	WECK	Type 6 NY Grip 11/16		
	34	"Ny-Grip" clamp 7/8	2 ea.	3223563-40	WECK	Type 6 NY Grip 7/8		
	35	"D" Washer	10 ea.	3223563-41	WECK	Type 191		GM5307-248-7477
	36	Stud, ¼-20 x 3 lg.	2 ea.	3223563-42	MALH	Dwg. 157		
	37	Screw, socket set ¼-20 x ¼ lg.	2 ea.	3223563-43			Stainless Steel TY-18-8	GM5305-227-1016
	38	Screw, flat head machine No. 10-32 x 1½ lg.	4 ea.	3223563-44			Stainless Steel TY-18-8	GM5305-274-7610
	39	Screw, Rd. Hd. Mach. No. 6-32 x 1¼ lg.	1 ea.	3223563-46			Stainless Steel TY-18-8	GS5305-010-0952
	40	Screw, Rd. Hd. Mach. No. 8-32 x 1 lg.	2 ea.	3223563-47			Brass	GM5305-290-3028
	41	Screw, Rd. Hd. Mach. No. 10-32 x 5/8 lg.	2 ea.	3223563-48			Stainless Steel TY-18-8	GS5305-010-1129
	42	Screw, Rd. Hd. Mach. No. 10-32 x 1 lg.	5 ea.	3223563-49			Stainless Steel TY-18-8	GM5305-290-2741
	43	Washer No. 6	4 ea.	3223563-50			Stainless Steel TY-18-8	GS5310-050-2257
	44	Washer No. 8	2 ea.	3223563-51			Brass, Nickel Plated	GS5310-050-2258
	45	Washer No. 10	4 ea.	3223563-52			Stainless Steel TY-18-8	GS5310-276-1195
	46	Washer, ¼	4 ea.	3223563-53			Brass	GM5310-194-7286
	47	Washer, Lock ¼	2 ea.	3223563-55			Stainless Steel	GM5310-187-2425
	48	Nut, self locking No. 6-32	5 ea.	3223563-56	EN	96TM-62		GA5310-194-8604
	49	Nut, self locking No. 8-32	2 ea.	3223563-57	EN	96TM-82		GA5310-194-8065
	50	Nut, self locking No. 10-32	15 ea.	3223563-58	EN	96TM-02		GA5310-263-2880
	51	Nut, self locking No. ¼-20	4 ea.	3223563-59	EN	96TU-04C		GA5310-188-1982
	52	Nut, self locking No. ¼-20	3 ea.	3223563-60	EN	96TE048		GF5310-199-6541
	53	Nut, Std. Hex ¼-20	2 ea.	3223563-61			Stainless Steel TY-18-8	GS5310-188-8942
	54	Screw, Rd. Hd. Mach. No. 6-32 x 7/16	2 ea.	3223563-62			Stainless Steel TY-18-8	GS5305-273-7337
	55	Screw, Rd. Hd. Mach. No. 10-32 x 5/8	2 ea.	3223563-63			Stainless Steel TY-18-8	GS5305-290-2181
	56	Screw, Rd. Hd. Mach. No. 6-32 x 11/16	4 ea.	3223563-64			Stainless Steel TY-18-8	GS5305-273-7340
	57	# 6 Lockwasher	4 ea.	3223563-65			Stainless Steel TY-18-8	GM5310-194-7310
	58	Wire No. 16 Stranded, Thermo-Plas. Red	61 ea.	3223563-66	BBR	7207-1-Red		GX6145-184-5351
	59	Wire No. 16 Stranded, Thermo-Plas. White	47 ea.	3223563-67	BBR	7207-1-White		GX6145-184-5348

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Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
	60	Wire No. 16 Stranded, Thermo-Plas. Blue	42 ea.	3223563-68	BBR	7207-1-Blue		GX6145-191-2574
	61	Wire No. 16 Stranded, Thermo-Plas. Green	75 ea.	3223563-69	BBR	7207-1-Green		GX6145-191-2571
	62	Wire No. 16 Stranded, Thermo-Plas. Yellow	84 ea.	3223563-70	BBR	7207-1 Yellow		GX6145-184-5345
	63	Wire No. 16 Stranded, Thermo-Plas. Black	235 ea.	3223563-71	BBR	7207-1 Black		GX6145-191-2577
	64	Wire No. 16 Stranded, Thermo-Plas. Orange	176 ea.	3223563-72	BBR	7207-1 Orange		GX6145-191-2568
	65	Sleeving 5/16 Dia. Black Vinyl Plas.	10 ea.	3223563-73	BBR			
	66	Sleeving 7/16 Dia. Black Vinyl Plas.	12 ea.	3223563-74	BBR			
	67	Sleeving ½ Dia. Black Vinyl Plas.	16 ea.	3223563-75	BBR			
	68	Sleeving ¾ Dia. Black Vinyl Plas.	35 ea.	3223563-76	BBR			
	69	#10 Black solid wire	40 ea.	3223563-77	GECM	AVA-SI-57046		
	70	#4 Stranded 1672/.0050 GE Deltabestos Power Cable	143 ea.	3223563-78	GECM	AA-SI-57048		
	71	Capacitor Hold Down	1 ea.	3223563-79	TTE	12T-Top Hat 42 Post		
	72	Washer, No. 6	4 ea.	3223563-80			Stainless Steel	GS5310-194-7331
	73	Screw, B.H.B. 6-32-3/16	4 ea.	3223563-81			Stainless Steel	GS5305-273-7333
	74	Lug 3/8 Plastic grip ring-Long	3 ea.	3223563-86	ARP			
	75	Lug 3/8 10 AWG Plastic Grip Ring tongue	2 ea.	3223563-87	ARP			
3-2	76	Screw #4 x 1/8" Lg	4 ea.	3223563-88			St. Drive	GT5305-253-5609
	77	Screw #6 x 3/16 lg	4 ea.	3223563-89			St. Drive	GS5305-253-5618
3-1	78	Washer, Lock 3/8	1 ea.					GF5310-260-7895
	79	Nut, Std. Hex 3/8-16	1 ea.					
	80	Plate Mounting	1 ea.	3223564-1	MALH			
	81	Resistor, Temp, COEF + .003% Per. Deg. C 200 ohm + ½%	1 ea.	3223564-2	MEP	RB15E	Part of full wave Rectifier	
	82	Resistor, Temp. COEF+.003% Per. Deg. C 25,000 ohm+½%	1 ea.	3223564-3	MEP	RB16E	Part of full wave Rectifier	

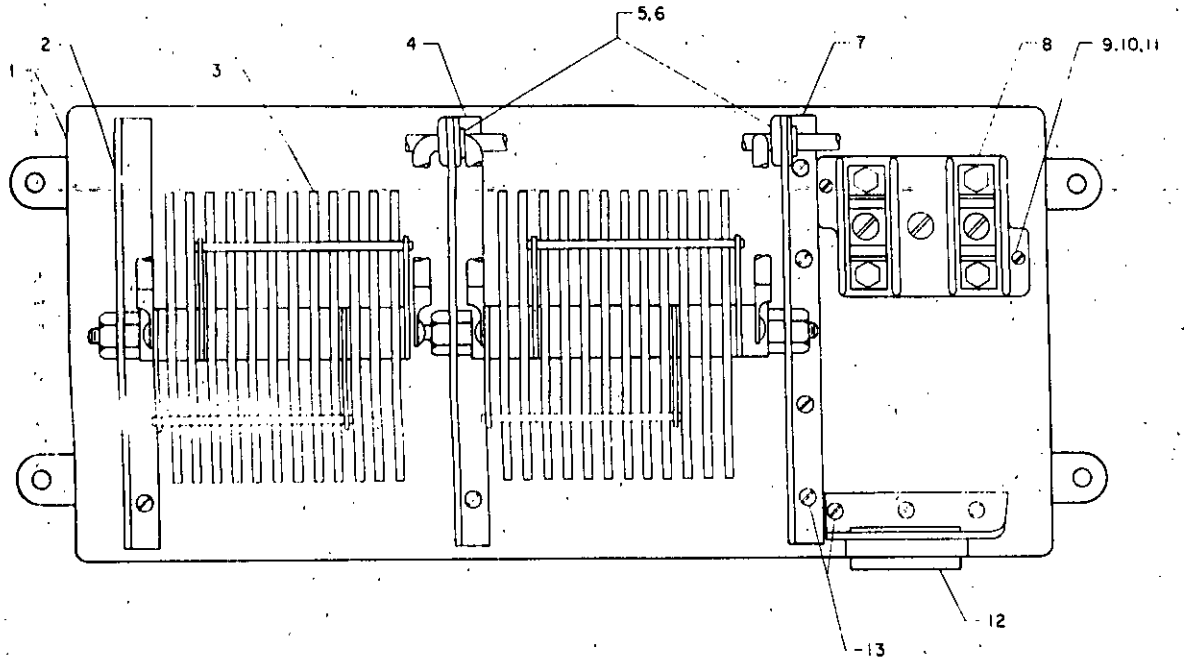
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Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
	83	Rectifier, Copper oxide	1 ea.	3223564-4	CNEL	TY-B-1	Part of full wave Rectifier	
	84	Milliammeter (Uncalibrated) (See Note 1)	1 ea.	3223564-5	GEMI	8DB14A3BH1	Indicates RPM of motor by measuring rectified tachometer output. (Calibrated by Meco)	
	85	Terminal, Solderless No. 8 hole	1 ea.	3223564-7	ARP	34145		
	86	Terminal, Solderless ¼ hole	2 ea.	3223564-8	ARP	31904		
	87	Screw, flat hd. brass mach. 5-40 x 1	2 ea.	3223564-10			Brass	GM5305-022-7349
	88	Screw, Rd. Hd. brass mach. 8-32 x ¾	2 ea.	3223564-12			Brass	GS5305-273-7359
	89	Nut, Brass Hex 5-40	2 ea.	3223564-14			Brass	GM5310-199-6547
	90	Nut, Brass Hex 8-32	2 ea.	3223564-15			Brass	GF5310-199-6538
	91	Nut, Lock 8-32	2 ea.	3223564-16	EN	96TM82		GM5310-199-6943
	92	Washer, Brass 1/8	2 ea.	3223564-18			Brass	GS5310-276-1191
	93	Washer, Brass 3/16	2 ea.	3223564-19			Brass	GS5310-050-2256
	94	Screw, Flat Hd. Brass Mach. 2-56 x 5/8	1 ea.	3223564-20			Brass	GS5305-022-7312
	95	Hex Brass Nut 2-56	1 ea.	3223564-21			Brass	GM5310-265-9628
	96	Washer # 8 Lock	2 ea.	3223564-22			Stainless Steel	GM5310-194-7352
3-2	97	Case, Control Cabinet	1 ea.	3223566-1	MALH			
	98	Cover, Control Cabinet	1 ea.	3223566-2	MALH			
	99	Panel, Instrument	1 ea.	3223566-3	MALH			
	100	Retainer, Gasket	1 ea.	3223566-4	MALH			
	101	Retainer, Guard	1 ea.	3223566-8	MALH			
	102	Guard, splashproof	1 ea.	3223566-9	MALH			
	103	Gasket, Meter	2 ea.	3223566-10	MALH			
	104	Gasket, Circuit Breaker	1 ea.	3223566-11	MALH			
	105	Seal, Rheostat	1 ea.	3223566-12	MALH			
	106	Handle, Clamp	4 ea.	3223566-13	MALH			
	107	Clamp, Panel	4 ea.	3223566-14	MALH			
	108	Clamp, Cover	4 ea.	3223566-15	MALH			
	109	Link, Cover clamp	8 ea.	3223566-16	MALH			
	110	Screw, CB Retaining	2 ea.	3223566-17	MALH			
	111	Pin, Clamp Handle	4 ea.	3223566-18	MALH			
	112	Pin, Cover Hinge	2 ea.	3223566-22	MALH			
	113	Gasket, Panel	1 ea.	3223566-24	MALH			
	114	Spring, Panel Hinge	4 ea.	3223566-25	MALH			

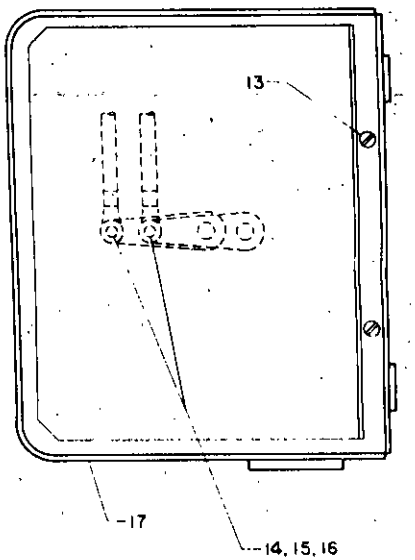
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Fig. No.	Pc. No.	Name and Description	No. Req.	Contractor Part No.	Manufacturer		Remarks	Navy Stock No.
					Name	Part No.		
	115	Gasket, Cover	1 ea.	3223566-28				
	116	O'Ring	4 ea.	3223566-34	PA	5427-7		
	117	O'Ring	1 ea.	3223566-35	PA	5430-20		
	118	Hex, Hd. Cap Scw. ¼-20 x 1	14 ea.	3223566-41			Stainless Steel	GM5305-174-4009
	119	Hex, Hd. Cap Scw. ¼-20 x 2½	2 ea.	3223566-42			Stainless Steel	GM5305-174-4015
	120	Flat Hd. Mach. Scw. 10-32 x 3/8	4 ea.	3223566-44			Stainless Steel	GS5305-290-2627
	121	Flat Hd. Mach. Scw. 10-32 x 9/16	16 ea.	3223566-45			Stainless Steel	GS 5305-290-2629
	122	Flat Hd. Mach. Scw. 10-24 x 3/4	2 ea.	3223566-46			Stainless Steel	GS5305-274-1109
	123	Allen Set Screw 3/8 x 24 x ½	4 ea.	3223566-47			Stainless Steel	GM5305-253-4796
	124	Lockwasher ¼ Std.	14 ea.	3223566-49			Stainless Steel	GM5310-194-7353
	125	'Palnut' 10-32	4 ea.	3223566-51	PALN	10M32		GA5310-262-6503
	126	Hex Nut ¼-20	6 ea.	3223566-52			Stainless Steel	GF5310-260-7922
	127	Elastic Stop Nut ¼-20	2 ea.	3223566-54	EN	96-T-U-040		GA5310-194-8639
	128	Rd. Hd. Rivet 3/16 x 7/8	8 ea.	3223566-55			Stainless Steel	GA5320-231-4039
	129	Split Cotter Pin 1/16 x ½	4 ea.	3223566-57			Stainless Steel	GM5315-234-1619
	130	Identification Plate	1 ea.	3223566-58	AMNP			
	131	Instruction Plate	1 ea.	3223566-59	AMNP			
	132	Instruction Sheet	1 ea.	3223566-60	MALH			
	133	Wiring diagram	1 ea.	3223566-61	MALH			
	134	Data sheet center holder	1 ea.	3223566-64	MALH			
	135	Nut, Lock 10-32	2 ea.	3223566-68	EN	96TM02		GF5310-199-6533
	136	Screw, Rd. Hd. Mach. No. 8-32 x 3/8	2 ea.	3223566-69			Brass	GM5305-281-3656

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PLAN VIEW (CAGE REMOVED)



END VIEW

Figure 3-3. Manual Controller, A Mark 6(b) NM MOD 1, Inverse Current Blocking Rectifier

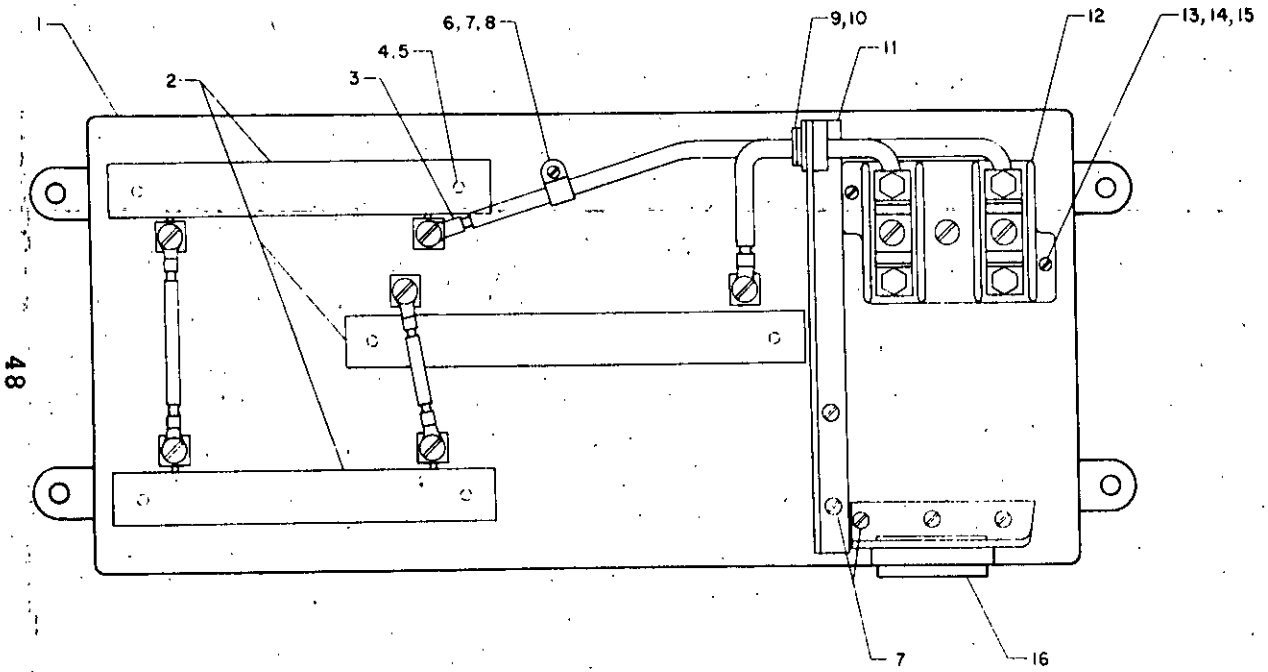
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TABLE 3-2

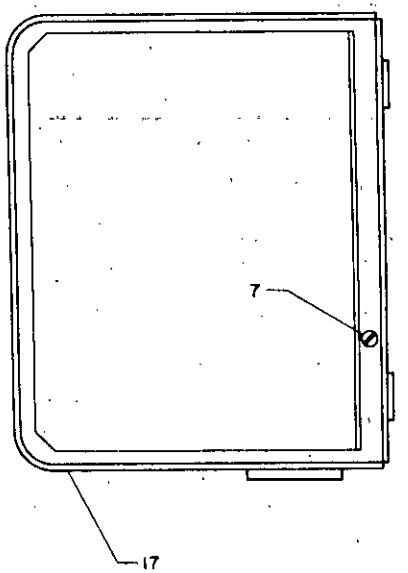
PARTS LIST, MANUAL CONTROLLER A MARK 6(b) NM MOD 1, INVERSE CURRENT BLOCKING RECTIFIER

Fig. & Index No.	Description	BuShips No.	No. Req.	Unit of Issue	Manufacturer's		Standard Navy Stock Number
					Name	Part No.	
3-3-	RECTIFIER, Inverse Current Blocking	C 3223552	Ref		COAE	11310	
-1	BASE, Rectifier Housing	C 3223553	1	each	COAE	11311	
-2	SUPPORT, Rectifier, rear	B 3223556	1	each	COAE	11314	
-3	RECTIFIER		2	each	KOTR	40H1J3S5M	
-4	SUPPORT, Rectifier, center	B 3223557	1	each	COAE	11315	
-5	BUSHING, Insulating, plastic, ¾ In. dia.		2	each	TOM	71M	
-6	NUT, Lock, No. 10-32		2	each	TOM		
-7	SUPPORT, Rectifier, front	B 3223558	1	each	COAE	11316	
-8	BLOCK, Terminal, 3 pole		1	each	SQ	Type U-3	
-9	SCREW, Mach, rd hd, stainless steel, No. 10-32 x 1 In. lg.		2	each			GS 5305-273-5944
-10	WASHER, Flat, stainless steel, No. 10		2	each			GM5310-187-4124
-11	NUT, Self-locking, No. 10-32		2	each	EN	96TM02	GA5310-188-2009
-12	BRACKET, Cable attachment	B 3223555	1	each	COAE	11313	
-13	SCREW, Mach, rd hd, stainless steel, No. 10-32 x		27	each			GS 5305-290-2177
-14	LUG, Terminal, Sta-Kon No. 6 AWG		4	each	TOM	E71	
-15	NUT, hex, stainless steel, No. 10-32		4	each			GF5310-199-6532
-16	WASHER, Lock, stainless steel, No. 10		4	each			GM5310-187-4124
-17	CAGE, Rectifier Housing	C 3223554	1	each	COAE	11312	

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PLAN VIEW (CAGE REMOVED)



END VIEW

Figure 3-4. Manual Controller, A Mark 6(b) NM MOD 1, Dynamic Braking Resistor

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TABLE 3-3

PARTS LIST, MANUAL CONTROLLER A MARK 6(b) NM MOD 1, DYNAMIC BRAKING RESISTOR

Fig & Index No.	Description	BuShips No.	No. Req.	Unit of Issue	Manufacturer's Name	Manufacturer's Part No.	Standard Navy Stock Number
3-4-	RESISTOR, DYNAMIC BRAKING	C 3223559	Ref		COAE	11324	
-1	BASE, Resistor Housing	C 3223560	1	each	COAE	11325	
-2	RESISTOR, 1.74 ohms at 22 Amp		3	each	SQ	Type N-92	
-3	LUG, Terminal, solderless		6	each	ARP	31127	
-4	SCREW, mach, rd hd, stainless steel, ¼-20 x ½ In. lg.		6	each			GS 5305-274-1126
-5	WASHER, Lock, stainless steel, ¼ in.		6	each			GM5310-194-7354
-6	CLAMP, "NY-LOCK", 3/16 In.		1	each	WECK	Type 3	
-7	SCREW, Mach, rd hd, stainless steel, No. 10-32 x ¼ In lg.		16	each			GS 5305-290-2177
-8	WASHER, "D"		1	each	WECK	No. 191	
-9	BUSHING, Insulating, plastic ½ In, dia.		1	each	TOM	70M	
-10	NUT, lock, ½ In.		1	each	TOM		
-11	BAFFLE	B 3223561	1	each	COAE	11326	
-12	BLOCK, Terminal, 3 pole		1	each	SQ	Type U-3	
-13	SCREW, Mach, rd hd, stainless steel, No. 10-32 x 1 In. lg.		2	each			GS 5305-273-5944
-14	WASHER, Flat, stainless steel, No. 10		2	each			GM5310-187-4124
-15	NUT, Self-locking, No. 10-32		2	each	EN	96TM02	GA5310-188-2009
-16	BRACKET, Cable attachment	B 3223555	1	each	COAE	11313	
-17	CAGE, Resistor Housing	C 3223554	1	each	COAE	11312	

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE
(NAVSHIPS OVPRT 12-66)

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